

BIOL 250 — Thorax Comprehensive Study Notes

Ch 24 — The Respiratory System (p. 624–649)

24.1 | An Overview of the Respiratory System and Respiratory Tract

- The respiratory system consists of the nose, nasal cavity, paranasal sinuses, pharynx, larynx (voice box), trachea (windpipe), and smaller conducting passageways down to the alveoli. Alveoli are the gas exchange surfaces within the lungs.
- Anatomical division: **upper respiratory system** = paranasal sinuses, nasal conchae, nose, nasal cavity, nasopharynx — these "condition the air" (filter, warm, humidify) to protect the lower respiratory system from debris, pathogens, and environmental extremes.
- **Lower respiratory system** = larynx, trachea, bronchi, lungs, bronchioles, and alveoli of the lungs.
- Functional division of the respiratory tract:
 - **Conducting portion:** extends from the entrance of the nasal cavity to the bronchioles; no gas exchange.
 - **Respiratory portion:** includes the respiratory bronchioles and the alveoli; gas exchange occurs here.
- By the time air reaches the alveoli, most foreign particles/pathogens have been removed and humidity/temperature are within acceptable limits — due to properties of the respiratory epithelium.
- **Functions of the respiratory system** (6 total):
 1. Provides an extensive area for gas exchange between air and circulating blood.
 2. Moves air to and from the exchange surfaces of the lungs.
 3. Protects respiratory surfaces from dehydration, temperature changes, and other environmental variations.
 4. Defends the respiratory system and other tissues from pathogens.
 5. Produces sounds involved in speaking, singing, nonverbal communication.
 6. Helps regulate blood volume, blood pressure, and body fluid pH.
 - Performs these functions along with the cardiovascular system, lymphatic system, selected skeletal muscles, and nervous system.

The Respiratory Epithelium

- **Respiratory epithelium** = pseudostratified ciliated columnar epithelium with numerous mucous cells; lines the entire respiratory tract EXCEPT the inferior portions of the pharynx, the smallest conducting passages, and the alveoli.
- A **stratified squamous epithelium** lines the inferior portion of the pharynx (protects from abrasion); this portion conducts air to the larynx and conveys food to the esophagus.
- Mucous cells in the epithelium and mucous glands in the lamina propria produce sticky **mucus** that bathes exposed surfaces.
- In the nasal cavity, cilia sweep mucus and trapped debris/microorganisms toward the pharynx to be swallowed (exposed to stomach acids/enzymes).
- In the lower respiratory system, cilia beat toward the pharynx, moving a carpet of mucus — this is the **mucociliary escalator**.
- **Respiratory defense system:** filtration mechanisms that remove contaminants.
 - In the nasal cavity, almost all particles larger than 10 μm are removed.

- **Vibrissae** = stiff hairs within the nasal cavity that remove larger particles.
- Mucus of the nasopharynx/pharynx traps smaller particles.
- Unpleasant stimuli (noxious vapors, dust, allergens, pathogens) rapidly increase mucus production rate.
- The common cold results when any of >200 types of viruses invade the respiratory epithelium.
- Filtration/warming/humidification occurs throughout the conducting portion, but the greatest changes occur in the nasal cavity. Mouth breathing eliminates much of this initial conditioning.
- Patients on a mechanical ventilator (bypasses nasal conditioning via tracheal tube) must receive externally filtered/humidified air to prevent alveolar damage.
- **Clinical correlation — Cystic Fibrosis (CF):** hereditary disorder of exocrine glands; mutation in a chloride regulator gene on chromosome 7 causes abnormally thick mucus production. Obstructs pancreatic/bile ducts, intestines, bronchial tree. In the respiratory tract, impedes breathing and impairs the mucociliary escalator, leading to frequent infections. Average lifespan for CF patients who reach adulthood is 37 years.

24.2 | The Upper Respiratory System

The Nose and Nasal Cavity

- Nose = primary passageway for incoming air. Air enters through paired **nostrils (nares)**, opening into the **nasal vestibule** and nasal cavity.
- **Nasal vestibule** = portion of nasal cavity enclosed by flexible tissues of the nose; supported by thin paired lateral and alar cartilages; epithelium contains coarse hairs across the external nares that trap large airborne particles (sand, sawdust, even insects).
- **Nasal septum** separates right/left nasal cavity; bony portion formed by fusion of the perpendicular plate of the ethmoid and the vomer; anterior portion is hyaline cartilage, which supports the dorsum (bridge) and apex (tip) of the nose.
- Lateral/superior walls of nasal cavity formed by maxillae, nasal and frontal bones, ethmoid, and sphenoid.
- Mucus from paranasal sinuses, aided by tears via the nasolacrimal ducts, keeps nasal cavity surface moist/clean.
- **Olfactory region** (superior nasal cavity) lined by olfactory epithelium: (1) inferior surface of the cribriform plate, (2) superior nasal conchae of the ethmoid, (3) superior portion of the nasal septum.
- **Superior, middle, inferior nasal conchae** (turbinated bones) project toward the septum from the lateral walls. Air flows between conchae and through the superior/middle/inferior **meatuses** as it passes from vestibule to the posterior nasal apertures (choanae).
- Incoming air bounces off conchae, creating turbulence (like water over rapids) — brings small particles into contact with mucus, allows extra time for warming/humidifying, and creates currents bringing olfactory stimuli to olfactory receptors.
- **Hard palate** (maxillae + palatine bones) forms floor of nasal cavity, separates oral/nasal cavities. **Soft palate** extends posterior to hard palate, marking the boundary between the superior nasopharynx and rest of pharynx.
- Nasal cavity opens into the nasopharynx at the posterior nasal apertures.

The Pharynx

- **Pharynx** connects nose, mouth, throat; shared by digestive and respiratory systems. Extends between posterior nasal apertures and the entrances to the trachea and esophagus.
- Curving superior/posterior walls attach to axial skeleton; lateral walls are flexible/muscular.
- Three regions: **nasopharynx, oropharynx, laryngopharynx**.
- **Nasopharynx:** superior portion of pharynx; connects to posterior nasal cavity via posterior nasal apertures; soft palate separates it from oral cavity. Lined by typical respiratory epithelium. **Pharyngeal tonsil** located on posterior wall; lateral walls contain the pharyngeal opening of the **auditory tube**.
- **Oropharynx:** extends between the soft palate and base of tongue at the level of the hyoid bone; communicates directly with the oral cavity. Lined with nonkeratinized stratified squamous epithelium at the nasopharynx/oropharynx boundary.

- Posterior margin of the soft palate supports the dangling **uvula** and two pairs of muscular pharyngeal arches (posterior arch, anterior arch).
- **Laryngopharynx**: narrow region between the hyoid bone and entrance to esophagus; lined with stratified squamous epithelium (resists abrasion, chemicals, pathogens).

24.3 | The Lower Respiratory System

The Larynx

- Air leaves the pharynx through the **glottis** (slit-like opening).
- **Larynx** begins between vertebra C3–C5 and ends at C6 or C7. Ligaments and skeletal muscles stabilize its cartilaginous walls.
- Three large unpaired cartilages form the body of the larynx: **thyroid cartilage**, **cricoid cartilage**, **epiglottis**. Thyroid and cricoid = hyaline cartilage; epiglottis = elastic cartilage.
- **Thyroid cartilage**: largest laryngeal cartilage ("shield-shaped"); forms most of anterior/lateral walls; incomplete posteriorly. **Laryngeal prominence (Adam's apple)** on anterior surface. Inferior surface articulates with cricoid cartilage; superior surface has ligamentous attachments to epiglottis and smaller laryngeal cartilages.
- **Cricoid cartilage** ("ring-shaped"): forms a complete circle; posterior portion greatly expanded, supports posterior larynx surface. Cricoid + thyroid cartilages protect the glottis and entrance to trachea. Inferior surface ligamentously attached to first tracheal cartilage; superior surface articulates with the small paired arytenoid cartilages.
- **Epiglottis**: shoehorn-shaped; projects with cartilage having ligamentous attachments to inner/superior borders of thyroid cartilage and hyoid bone. During swallowing, larynx elevates and epiglottis folds back over the glottis, preventing liquids/solids from entering airway.
- **Paired laryngeal cartilages** (3 pairs): **arytenoid**, **corniculate**, **cuneiform**. Arytenoids and corniculates = hyaline cartilage; cuneiforms = elastic cartilage.
 - **Arytenoid** ("ladle-shaped") cartilages articulate with posterior/superior border of the cricoid cartilage.
 - **Corniculate** ("horn-shaped") cartilages articulate with the arytenoid cartilages; important in opening/closing the glottis and producing sound.
 - **Cuneiform** ("wedge-shaped") cartilages lie within the ary-epiglottic fold (tissue between lateral arytenoid and epiglottis).
- **Laryngeal ligaments**: intrinsic ligaments bind all nine cartilages together. Extrinsic ligaments attach thyroid cartilage to hyoid bone and cricoid cartilage to trachea. **Vestibular ligaments** and **vocal ligaments** extend between the thyroid cartilage and the arytenoids.
- Folds of laryngeal epithelium project into the glottis, covering these ligaments:
 - **Vestibular folds** (superior pair) = "false vocal cords"; relatively inelastic; prevent foreign objects from entering the glottis, protect the vocal folds; play no role in sound production.
 - **Vocal folds** = "true vocal cords"; contain the elastic vocal ligament; highly elastic; involved in sound production.
- **Sound production**: air passing through the glottis vibrates the vocal folds, producing sound waves. Pitch depends on diameter, length, and tension of vocal folds. Diameter/length relate to larynx size; tension controlled by skeletal muscle contraction that changes the distance between thyroid and arytenoid cartilages — increased distance = tensed folds = higher pitch; decreased distance = relaxed folds = lower pitch.
 - Children: slender, short vocal folds → high-pitched voices.
 - At puberty, male larynx enlarges more than female; adult male true vocal cords are thicker/longer, producing lower tones.
 - Entire larynx vibrates, creating composite sound; amplification/echoing occurs in the pharynx, oral cavity, nasal cavity, paranasal sinuses. Final distinct sound production depends on voluntary movements of tongue, lips, cheeks.
- **Laryngeal musculature**: two groups —

- **Intrinsic laryngeal muscles:** entirely within the larynx; two functions — (1) regulate tension in vocal folds (insert on thyroid, arytenoid, corniculate cartilages), (2) open/close the glottis (insert onto arytenoid cartilages).
- **Extrinsic laryngeal muscles:** connect larynx to neighboring structures; move it up/down during swallowing and speaking (covered in Ch10).
- Extrinsic + intrinsic muscles cooperate to prevent food/liquid from entering the glottis during swallowing: chewed food forms a **bolus**; extrinsic muscles elevate the larynx, bending the epiglottis over the glottis entrance so the bolus glides over the epiglottis rather than falling in; intrinsic muscles simultaneously close the glottis. If particles/liquids touch vestibular or vocal folds, the **coughing reflex** is triggered to prevent material from entering the glottis.
- **Clinical note — Tracheal Blockage:** Foreign objects (usually food) lodged in trachea/larynx are usually expelled by coughing. If a person can speak, airway is open, no emergency measures needed. If person cannot breathe/speak but is conscious → life-threatening emergency.
 - **Heimlich maneuver:** expels tracheal obstruction via abdominal compression just inferior to the diaphragm — grasp fist from behind, thrust inward/upward to force diaphragm upward and air into trachea to dislodge object.
 - **Tracheostomy:** incision through anterior tracheal wall, tube inserted inferior to larynx; performed by qualified rescuers if blockage remains. Also required if: (1) larynx blocked by inflammation, laryngeal spasms, severe allergic reactions, burns, smoke inhalation; (2) trachea crushed; (3) epiglottis severely inflamed; (4) part of trachea removed during laryngeal cancer treatment.

24.4 | The Trachea

- **Trachea** begins anterior to vertebra C6, attached to the cricoid cartilage by connective tissue; ends within the mediastinum at T5, where it branches into right and left main bronchi.
- Lining: pseudostratified ciliated columnar epithelium (respiratory epithelium) with mucous cells, sitting on the **lamina propria** (loose connective tissue), which separates the epithelium from deeper hyaline cartilages. Epithelium + lamina propria = **mucosa**.
- **Submucosa:** connective tissue layer deep to lamina propria; contains **tracheal glands** that secrete mucus onto the epithelial surface via secretory ducts.
- Deep to submucosa: **15–20 tracheal cartilages**; elastic **annular ligaments** attach each cartilage to adjacent ones. Tracheal cartilages stiffen tracheal walls, protect the airway, and prevent collapse/overexpansion as pressures change.
- Each tracheal cartilage is a **C-shaped** piece of hyaline cartilage. Closed portion of the C protects anterior/lateral surfaces; open portions face posteriorly toward the esophagus — because not complete circles, the posterior tracheal wall can change shape easily during swallowing, permitting passage of large food masses along the esophagus.
- **Trachealis:** elastic ligament + band of smooth muscle connecting the open ends of each tracheal cartilage. Contraction decreases tracheal lumen diameter, increasing airflow resistance. Sympathetic activation relaxes the trachealis, increasing diameter and lowering airflow resistance (easier movement of large air volumes).

24.5 | The Main Bronchi

- At **T5**, the trachea branches within the mediastinum into the right and left **main bronchi**.
- **Carina** of trachea: internal ridge lying between the entrances to the two main bronchi.
- Histology of main bronchi = same as trachea.
- Right main bronchus supplies the right lung; left supplies left lung.
- Right main bronchus is shorter, larger diameter, and descends at a steeper angle than the left → most foreign objects entering the trachea end up in the right bronchus.
- Each main bronchus travels to/through the **hilum** of the lung — the site of entry/exit for pulmonary vessels and nerves.

- **Root of the lung:** meshwork of dense connective tissue anchoring the bronchus, vessels, nerves, lymphatics to the lung; attached to the mediastinum; holds these structures in place.
- Roots of the lungs are located anterior to vertebrae T5 (right) and T6 (left).

24.6 | The Lungs

Gross Anatomy / Lobes and Fissures

- Each lung is cone-shaped: **apex** extends superior to the attachment of the first rib to the sternum; **base** sits on the diaphragm.
- Lungs occupy the left and right pleural cavities.
- **Right lung:** 3 lobes (superior, middle, inferior). **Horizontal fissure** separates superior/middle; **oblique fissure** separates superior/inferior (technically separates inferior from sup+middle).
- **Left lung:** 2 lobes (superior, inferior), separated by the oblique fissure.
- Right lung is shorter, broader, heavier than left. Shorter because the right half of the diaphragm rises higher to accommodate the liver. Left lung is narrower/lighter because the heart and great vessels project into the left pleural cavity.

Lung Surfaces

- **Costal surface:** curving anterior surface, contacts inner contours of the rib cage.
- **Mediastinal surface:** contains the hilum of the lung; bears grooves marking the great vessels and heart.
- Left lung has a larger **cardiac impression** (most of heart is left of midline).
- Right lung's medial margin forms a vertical line anteriorly; left lung has an indentation, the **cardiac notch**.
- Root connective tissue extends inward forming **trabeculae** (branch repeatedly; contain elastic fibers, smooth muscle, lymphatic vessels), dividing lobes into progressively smaller compartments. Conducting passageways, pulmonary vessels, nerves follow trabeculae to peripheral destinations.
- Terminal connective tissue partitions = **interlobular septa**, dividing the lung into **pulmonary lobules**. Each lobule supplied by branches of pulmonary arteries, pulmonary veins, and respiratory passageways. Septa connective tissue is continuous with the visceral pleura.

Bronchi and Bronchioles / Bronchial Tree

- Main bronchi + branches = **bronchial tree**. An airway branches ~23 times before reaching the level of a lobule.
- Each main bronchus divides into **lobar (secondary) bronchi**, which divide into **segmental (tertiary) bronchi**. Branching pattern differs between right/left lungs.
- Each segmental bronchus supplies a single **bronchopulmonary segment**.
 - Right lung: 10 segmental bronchi, 10 bronchopulmonary segments.
 - Left lung (fetal): starts with 10 bronchopulmonary segments, but some fuse during development → adult left lung has 8 or 9 bronchopulmonary segments (text also states "usually nine").
- Walls of main, lobar, segmental bronchi have typical respiratory epithelium but contain progressively less cartilage with each division. Lobar/segmental bronchi walls contain cartilage plates around the lumen (same function as tracheal cartilage rings).
- **Right main bronchus** divides into 3 lobar bronchi: superior, middle, inferior — each delivers air to one lobe.
- **Left main bronchus** divides into 2 branches: superior lobar bronchus, inferior lobar bronchus.
 - Right lung: 10 segmental bronchi — 3 supply superior lobe, 2 supply middle lobe, 5 supply inferior lobe.
 - Left lung: usually 9 segmental bronchi — 4 supply superior lobe, 5 supply inferior lobe.
- **Bronchopulmonary segments:** each is the lung tissue associated with a single segmental bronchus; named to match the associated segmental bronchus.

Bronchioles

- Each segmental bronchus branches multiple times within its bronchopulmonary segment, forming many bronchioles, which branch into the smallest-diameter conducting branches: **terminal bronchioles**.
- ~6500 terminal bronchioles arise from each segmental bronchus.

- Terminal bronchiole walls lack cartilage and mucous cells; proximal segments lined by pseudostratified ciliated columnar epithelium; more distal segments lined by simple columnar ciliated epithelium.
- Terminal bronchiole walls contain significant smooth muscle, regulated by the ANS:
 - **Sympathetic activation** + epinephrine release from adrenal medullae → **bronchodilation** (increases luminal diameter, reduces airflow resistance).
 - **Parasympathetic stimulation** → **bronchoconstriction** (reduces luminal diameter, increases airflow resistance). Excessive parasympathetic stimulation (as in asthma) can almost completely prevent airflow along terminal bronchioles.
- Each terminal bronchiole delivers air to a single **pulmonary lobule**.
- Within the lobule, terminal bronchiole branches into several **respiratory bronchioles** — thinnest, most delicate branches of the bronchial tree; incomplete walls, lined by simple cuboidal epithelium lacking cilia.
- **Clinical note — Lung Cancer:** aggressive malignancy class; kills more people than any other cancer type. Affects epithelial cells of conducting passageways, mucous glands, or alveoli. Symptoms (chest pain, shortness of breath, cough/wheeze, weight loss) usually don't appear until tumors restrict airflow or compress structures. 85–90% of lung cancers result directly from cigarette smoking. Five-year survival: 54% if localized at detection, but only 15% of lung cancers are diagnosed at this early stage; <5% five-year survival if metastasized.

Alveolar Ducts and Alveoli

- Respiratory bronchioles connect to individual alveoli and to multiple alveoli along **alveolar ducts**. Alveolar ducts end at **alveolar sacs** — common chambers connected to multiple alveoli.
- Each lung contains ~150 million alveoli, giving lungs an open, spongy appearance.
- Alveolar ducts lined with simple squamous epithelium; isolated smooth muscle cells in interalveolar septa; extensive capillary network surrounds each alveolus, surrounded by elastic fiber network. Elastic tissue maintains positions of alveoli/ducts/bronchioles; stretches during inspiration, recoils during expiration (reduces alveolar size, assists expiration).

The Alveolus and the Blood Air Barrier

- Alveolar cell layer = mainly simple squamous epithelium.
- **Type I alveolar cells:** unusually thin squamous cells; sites of gas diffusion.
- **Alveolar macrophages:** roam and patrol the epithelial surface, engulfing particles that eluded other defenses.
- **Type II alveolar cells:** large cells scattered among squamous cells; produce **surfactant** — an oily secretion containing phospholipids and proteins. Surfactant reduces surface tension by interacting with water molecules, preventing alveolar collapse, keeping alveoli open.
- **Blood-air barrier:** three layers — (1) alveolar cell layer, (2) capillary endothelial layer, (3) fused basement membrane between them.
- Gas exchange is rapid/efficient: very short diffusion distance; O₂ and CO₂ are small, lipid-soluble molecules; surface area of the blood-air barrier is very large.
- **Clinical note — Neonatal Respiratory Distress Syndrome (RDS):** breathing disorder of premature newborns where alveoli fail to remain open due to absent/insufficient surfactant. Surfactant production begins at end of 6th month of fetal development; by 8th month reaches adequate levels. Without enough surfactant, alveoli collapse on exhalation and stick together, forcing extra-forceful inhalation; infant exhausts quickly, gas exchange ineffective. Risk reduced by delaying delivery until fetal lungs develop. Most cases occur in babies born before 37–39 weeks. Other risk factors: problems with blood flow during delivery, multiple pregnancy, rapid labor, maternal diabetes.
 - Treatment: **CPAP** (continuous positive airway pressure) — delivers air under pressure to hold alveoli open, keeping newborn alive until surfactant production normalizes. Also: administering synthetic surfactant via endotracheal tube immediately after delivery and repeatedly over first days of life.
- **Clinical note — Adult Respiratory Distress Syndrome (ARDS):** caused by inadequate surfactant production leading to alveolar collapse on exhalation; results from severe respiratory infections or pulmonary injuries. Treated with **PEEP** (positive end-expiratory pressure) — airway pressure greater than atmospheric maintained

at end of exhalation via device. PEEP maintains life until underlying problem heals; ARDS has a high mortality rate.

The Blood Supply to the Lungs

- Lungs receive blood from two sources: the **pulmonary circuit** and the **bronchial arteries** of the systemic circuit.
- Respiratory exchange surfaces receive blood from the pulmonary circuit arteries. **Pulmonary arteries** enter the lungs at the hilum, branch with the bronchi toward the lobules. Each lobule receives an arteriole; a venule and capillary network surround each alveolus as part of the blood-air barrier. Blood from alveolar capillaries → pulmonary venules → pulmonary veins → left atrium.
- Endothelial cells of alveolar capillaries are the primary source of **angiotensin-converting enzyme (ACE)**, which converts circulating angiotensin I to angiotensin II — important in regulating blood volume and blood pressure.
- Nasal passages and larynx (conducting portion) receive blood from the **external carotid arteries**.
- **Thyrocervical trunks** (branches of subclavian arteries) supply blood to the inferior larynx and trachea.
- **Bronchial arteries** supply blood to the bronchial tree and alveoli; venous blood from these structures flows into the pulmonary veins, bypassing the rest of the systemic circuit and diluting oxygen-rich blood leaving the alveoli.
- **Clinical note — COPD (Chronic Obstructive Pulmonary Disease):** group of disorders restricting airflow and reducing alveolar ventilation. Types: asthma, chronic bronchitis, emphysema.
 - **Asthma:** inflammatory lung disease — bronchial inflammation, spasms, difficulty breathing. Can result from hypersensitivity/allergy, cold temperature, or exercise. During an attack: smooth muscle spasms constrict airways + excessive mucus production. Symptoms: coughing, wheezing, shortness of breath, chest tightness (like "breathing through a small straw"). Allergic asthma = reaction to environmental allergens (animal dander, dust mites, pollen, mold); non-allergic triggers = smoke, pollution, cold air; exercise-induced asthma = response to physical activity. Majority of children with asthma develop it before age 5. No cure, but early diagnosis/aggressive treatment manages symptoms.
 - **Chronic bronchitis:** recurrent, long-term inflammation/swelling of the bronchi → recurrent infections, excessive mucus, harsh cough with expectoration. Diagnostic criteria: symptoms on most days for >3 months of the year, for ≥2 consecutive years. Most common cause: cigarette smoking. Causes low blood oxygenation, cyanosis (bluish skin), widespread inflammation → "blue bloaters."
 - **Emphysema:** damaged, permanently enlarged alveoli causing breathlessness; lung resembles "an old, used sponge." Destruction of alveolar surfaces → inadequate gas-exchange surface area. Smooth muscles/thickened alveoli merge into large sacs; loss of lung elasticity traps air during expiration. Patients use much energy to breathe but maintain near-normal blood oxygenation, skin usually pink → "pink puffers." Main cause: smoking.

24.7 | The Pleural Cavities and Pleural Membranes

- Thoracic cavity = broad cone shape; walls = rib cage; floor = muscular diaphragm.
- Two pleural cavities separated by the **mediastinum**.
- Each lung occupies a single pleural cavity, lined by a serous membrane called **pleura**.
- Pleural membrane = two continuous layers:
 - **Parietal pleura:** covers inner surface of thoracic wall, superior surface of diaphragm, surfaces of the mediastinum.
 - **Visceral pleura:** covers outer surfaces of the lungs, extends into the fissures between lobes.
- **Pleural cavity:** space between parietal and visceral pleurae; a "potential space" rather than an open chamber (layers in close contact).
- Both pleural membranes secrete **pleural fluid** — moist, slippery coating that lubricates and reduces friction between parietal/visceral surfaces during breathing.

- **Clinical correlation — Pleurisy:** inflammation of the pleurae; may (1) cause membranes to produce/secrete excess pleural fluid, or (2) cause inflamed pleurae to adhere to one another, limiting movement. Either way, breathing becomes difficult, requiring prompt medical attention.

24.8 | Respiratory Muscles and Pulmonary Ventilation

- **Pulmonary ventilation** (breathing) = physically moving air into/out of the lungs. Its function: maintain adequate **alveolar ventilation** — movement of air into/out of alveoli — preventing CO₂ buildup and ensuring continual O₂ supply matching bloodstream absorption.

Respiratory Muscles

- Most important muscles: **diaphragm** and **external intercostals** — classified as **primary respiratory muscles**, active during normal breathing at rest.
- **Accessory respiratory muscles** become active when depth/rate of breathing must increase markedly (e.g., exercise). Accessory muscles include: sternocleidomastoid, serratus anterior, transversus thoracis, scalenes, pectoralis minor, external oblique, rectus abdominis (and internal intercostals contribute to forced exhalation).
- Respiratory movements classified as **eupnea** (quiet/normal breathing, passive expiration) or **hyperpnea** (forced breathing, active expiration).
- **Eupnea:** lung expansion stretches elastic fibers; rib cage elevation stretches skeletal muscles/elastic fibers of body wall. When inspiratory muscles relax, elastic recoil returns diaphragm/rib cage to original position (passive exhalation).
 - **Diaphragmatic breathing** (deep breathing): diaphragm contraction changes thoracic volume; air drawn in as diaphragm contracts, exhalation as it relaxes.
 - **Costal breathing** (shallow breathing): thoracic volume changes because rib cage changes shape. Inhalation = external intercostal contraction elevates ribs, enlarges thoracic cavity; exhalation = these muscles relax. Pregnant women increasingly rely on costal breathing as the uterus enlarges and pushes abdominal viscera against the diaphragm.
- **Hyperpnea:** accessory muscles assist inspiration; expiration involves contraction of the **transversus thoracis** and **internal intercostal muscles**. At maximum breathing levels (e.g., vigorous exercise), abdominal muscles also assist exhalation by compressing abdominal contents against the diaphragm, reducing thoracic volume.
- Rib elevation during inhalation resembles the outward swing of a raised bucket handle.
- Summary of muscle roles (Spotlight figure):
 - Primary inhalation muscles: diaphragm, external intercostals.
 - Accessory inhalation muscles: sternocleidomastoid, serratus anterior, pectoralis minor, scalenes, external oblique.
 - Exhalation (forced): internal intercostals, transversus thoracis, plus abdominal muscles compressing viscera up against diaphragm.

Reflex control of respiration

- Normal breathing occurs automatically without conscious control.
- Three reflex types regulate respiration:
 1. **Mechanoreceptor reflexes:** respond to changes in lung volume or arterial blood pressure.
 2. **Chemoreceptor reflexes:** respond to changes in PCO₂, pH, and PO₂ of blood and cerebrospinal fluid.
 3. **Protective reflexes:** respond to physical injury or irritation of the respiratory tract.
- Higher centers (cerebrum/cerebral cortex, hypothalamus) influence respiration via connections to the pneumotaxic center and direct influence on respiratory muscles. Pyramidal output = conscious control; most instructions actually arrive via extrapyramidal pathways. Respiratory centers embedded in the reticular formation connect with nearly every sensory/motor nucleus, so emotional and autonomic activities also affect respiration's pace/depth.

24.9 | Respiratory Changes at Birth

- Before delivery: pulmonary arterial resistance is high (pulmonary vessels collapsed); rib cage compressed; lungs/conducting passageways contain only small amounts of fluid, no air.
- At birth: newborn's first breath results from powerful contractions of the diaphragm and external intercostal muscles. Inspired air enters with enough force to push fetal fluids out of the way and inflate the entire bronchial tree and most alveoli; causes the **ductus arteriosus** to close.
- Newborn's first exhalation fails to completely empty the lungs because the rib cage doesn't return to its fully compressed state. Supporting cartilages/connective tissues keep passageways open, and surfactant on alveolar surfaces prevents their collapse. Subsequent breaths complete alveolar inflation.

24.10 | Respiratory Centers of the Brain

- Under normal conditions, O₂ absorption and CO₂ generation are matched via rates of O₂ delivery/CO₂ removal by capillary beds. When unequal, cardiovascular/respiratory systems make coordinated adjustments.
- Regulatory centers located in the **pons** and **medulla oblongata**.
- **Respiratory centers**: three pairs of loosely organized nuclei in the reticular formation of the pons and medulla oblongata; regulate respiratory muscle activity by adjusting frequency/depth of ventilation.
- **Respiratory rhythmicity centers**: set basic pace/depth of respiration; subdivided into:
 - **Dorsal respiratory group (DRG)** = inspiratory center: controls motor neurons innervating the external intercostal muscles and diaphragm; functions in every respiratory cycle (quiet or forced).
 - **Ventral respiratory group (VRG)**: functions only during forced respiration; innervates motor neurons controlling accessory muscles involved in active exhalation and maximal inhalation. Neurons for active exhalation sometimes called the "expiratory center."
- **Apneustic center** and **pneumotaxic center** (paired nuclei of the pons): adjust the output of the rhythmicity centers, modifying the pace of respiration in response to sensory stimuli or higher-center instructions.
 - Apneustic center: causes strong, sustained inspiratory movements.
 - Pneumotaxic center: inhibits the apneustic center and the inspiratory (DRG) center of the medulla.
- Sensory inputs to the respiratory control system: chemoreceptors and baroreceptors of the carotid and aortic sinuses (via cranial nerves IX and X), chemoreceptors responding to cerebrospinal fluid (CSF) composition, and inputs from the limbic system/hypothalamus/cerebrum.

24.11 | Aging and the Respiratory System

- Aging reduces respiratory efficiency via 4 notable factors:
1. Elastic tissue throughout the body deteriorates with age, reducing lungs' ability to inflate/deflate.
 2. Arthritic changes in rib joints and decreased flexibility of costal cartilages restrict rib cage movement.
 3. Stiffening/reduced chest movement limits respiratory volume, reducing exercise performance/capabilities.
 4. Some degree of emphysema appears in many individuals after age 50. On average, ~1 square foot of the blood-air barrier is lost each year after age 30 (varies widely with lifetime exposure to tobacco smoke and other respiratory irritants).

Clinical Case Wrap-Up & Related Clinical Terms (Ch 24)

- Coughs from colds/flu tend to last about **18 days** (not just ~1 week).
- Antibiotics only work against bacteria, not viruses — viruses cause **90%** of acute upper respiratory infections.
- Fever = innate immunity at work; excessive mucus production (productive cough, runny nose) indicates the immune system is working; irritates trachea/bronchi, causing parasympathetic reflexive coughing that clears viruses (but can also spread them to others).
- Related clinical terms:
 - **Aspirate**: to draw fluid from the body by suction, or foreign material accidentally sucked into the lungs.

- **Bronchography:** procedure introducing radiopaque materials into airways to improve x-ray imaging of the bronchial tree.
- **Cheyne-Stokes breathing:** hyperpnea (deep, fast breathing) alternating with apnea (absence of breathing).
- **Dyspnea:** condition of labored breathing.
- **Endotracheal tube:** tube passed through mouth/nose into the trachea.
- **Hemothorax:** condition of having blood in the pleural cavity.
- **Hyperbaric oxygenation:** therapy to force more oxygen into the blood using a pressure chamber.
- **Orthopnea:** condition in which one has difficulty breathing except when upright.
- **Rhinoplasty:** plastic surgery of the nose.
- **Sputum:** mixture of saliva and mucus coughed up from the respiratory tract, often due to infection.
- **Wheeze:** audible whistling sound when breathing.

Ch 21 — The Heart (p. 545–566)

21.1 | An Overview of the Cardiovascular System

- The heart is roughly the size of a fist; has four muscular chambers: right/left **atria** and right/left **ventricles**.
- Heart beats approximately **100,000 times/day**; pumps more than **1.5 million gallons of blood/year** (enough to fill 200 train tank cars). Equivalent to running a full-open kitchen faucet for **45 years** to match a lifetime's pumped blood volume.
- Cardiac output ranges from **5 to 30 liters per minute** depending on nervous system regulation and activity level.
- Vessel network divided into two circuits:
 - **Pulmonary circuit:** carries CO₂-rich blood from heart to lung gas-exchange surfaces, returns O₂-rich blood to heart.
 - **Systemic circuit:** transports O₂-rich blood from heart to body cells, returns CO₂-rich blood to heart.
- Right atrium receives blood from systemic circuit; right ventricle pumps into pulmonary circuit. Left atrium receives blood from pulmonary circuit; left ventricle pumps into systemic circuit.
- With each heartbeat: atria contract first, then ventricles; the two ventricles contract simultaneously, ejecting equal blood volumes into pulmonary and systemic circuits.
- Circuits function in sequence — blood from systemic circuit must complete the pulmonary circuit before re-entering systemic circuit.
- Vessel types in both circuits: arteries (carry blood away from heart), veins (return blood to heart), capillaries (small thin-walled vessels connecting smallest arteries/veins).

21.2 | The Pericardium

- Heart is located near the anterior chest wall, directly posterior to the sternum.
- The mediastinum contains the great vessels (attached at the heart's base), plus the thymus, esophagus, trachea.
- **Pericardium:** surrounds the heart; composed of two parts — outer **fibrous pericardium** and inner **serous pericardium**.
- Trachea located within the middle mediastinum, immediately posterior to the sternum; heart is anterior part of the mediastinum, immediately posterior to sternum.
- **Fibrous pericardium:** dense network of collagen fibers; stabilizes heart/vessel position within the mediastinum.
- **Serous pericardium:** two-layered membrane — outer **parietal layer** and inner **visceral layer** (the visceral layer = **epicardium**).
- **Pericardial cavity:** potential, fluid-filled space between the two serous layers.

- Contains up to **50 mL of pericardial fluid**, secreted by pericardial membranes; acts as lubricant reducing friction between visceral/parietal surfaces during heartbeats.
- Analogy: pushing a fist into a partially inflated balloon — balloon = pericardium, fist = heart.
- **Clinical correlation — Pericarditis**: infection/inflammation of the pericardium. Most frequent cause = viral infection; bacterial pericarditis can complicate tuberculosis. Other causes: cancer invading the pericardium, kidney failure (uremic pericarditis), severe chest trauma with bleeding into the pericardial sac.

21.3 | Structure of the Heart Wall

- Heart wall has three distinct layers: (1) outer **epicardium**, (2) middle **myocardium**, (3) inner **endocardium**.

1. **Epicardium** = visceral layer of serous pericardium covering the heart surface; has a mesothelium + underlying areolar tissue supporting layer. (Parietal layer of serous pericardium = outer dense fibrous layer + inner mesothelium.)

2. **Myocardium** = cardiac muscle tissue forming atria/ventricles, with associated connective tissue, blood vessels, nerves. Atrial myocardium is thin, organized in figure-eight layers passing between atria. Ventricular myocardium is much thicker; muscle orientation changes layer to layer — superficial muscles wrap around both ventricles, deeper layers spiral around/between individual ventricles.

3. **Endocardium** = simple squamous epithelium covering inner heart surfaces, including valves; continuous with the endothelium of attached great vessels.

Cardiac Muscle Tissue

- Cardiac muscle cells: small, averaging **10–20 µm diameter, 50–100 µm length**; single centrally placed nucleus.
- Resemble skeletal muscle fibers (organized myofibrils, striations) but differ in key respects:
 - Almost totally dependent on **aerobic respiration**; sarcoplasm contains hundreds of mitochondria and large myoglobin reserves (oxygen storage), plus large glycogen and lipid inclusions as energy reserves.
 - T tubules are shorter than in skeletal muscle and do NOT form triads with the sarcoplasmic reticulum.
 - Larger number of blood vessels than even red skeletal muscle tissue.
 - Contract without nervous system stimulation (autorhythmic).
 - Cells interconnected by specialized junctions called **intercalated discs**.

The Intercalated Discs

- Unique membranes give intercalated discs a jagged appearance. Features:
 - **Desmosomes**: bind plasma membranes of two cardiac cells together, locking them together, preventing separation during contraction.
 - **Fascia adherens**: specialized junction where actin filaments anchor firmly to the plasma membrane within the intercalated disc, tying together actin filaments of adjacent cells so the two cells "pull together" with maximum efficiency.
 - **Gap junctions** (communicating junctions): allow ions/small molecules to move between cells, creating direct electrical connection — action potentials move from cell to cell as if the sarcolemmae were continuous.
- Because cardiac muscle cells are mechanically, chemically, and electrically connected, cardiac muscle tissue functions like a single, enormous cell — contraction of one cell triggers contraction of others, spreading through the myocardium. Termed a **functional syncytium**.

Cardiac Skeleton

- Connective tissues of the heart include large numbers of reticular, collagen, and elastic fibers. Each cardiac muscle cell wrapped in an elastic sheath; adjacent cells tied by fibrous cross-links ("struts"). Muscle layers have fibrous wrapping; fibrous sheets separate superficial/deep layers.
- These connective tissue layers are continuous with dense connective tissue bands that: (1) encircle the bases of the pulmonary trunk and aorta, (2) encircle the heart valves, (3) connect the fibrous rings surrounding valve

openings, (4) extend into the cardiac muscle separating atria and ventricles. This network = the **cardiac skeleton**.

- Functions of the cardiac skeleton:
 - Stabilizes positions of muscle cells and valves.
 - Provides physical support for cardiac muscle cells, blood vessels, nerves within myocardium.
 - Distributes the forces of contraction.
 - Reinforces valves, helps prevent heart overexpansion.
 - Provides elasticity that returns the heart to original shape after each contraction.
 - Physically isolates atrial muscle cells from ventricular muscle cells (vital for coordinating cardiac contractions).

21.4 | Orientation and Superficial Anatomy of the Heart

- A midsagittal section does not cut the heart in half because the heart: (1) lies slightly left of midline, (2) sits at an angle to the body's longitudinal axis, (3) is rotated to the left.
- Heart located within the mediastinum between the two lungs. Because heart lies left of midline, the cardiac notch in the left lung is deeper than in the right lung.
- **Base of the heart:** broad superior portion, attached to major arteries/veins of systemic and pulmonary circuits; begins at the origins of major vessels and superior surfaces of the atria. Sits posterior to the sternum, approximately at the third costal cartilage. (Balloon analogy: base = your wrist.)
- **Apex of the heart:** inferior, pointed tip, formed mainly by the left ventricle; points laterally; reaches the fifth intercostal space, extends left of midline.
- Because heart is rotated slightly left: base = superior border; right atrium forms the right border; left ventricle + small portion of left atrium form the left border (extends to apex); apex formed mainly by left ventricle; inferior border formed mainly by inferior wall of right ventricle.
- **Anterior (sternocostal) surface:** faces anterior thoracic wall; consists mostly of right ventricle wall (~2/3) and some left ventricle (~1/3).
- **Posterior surface** (at the base): formed by the left atrium and a small portion of the right atrium.
- **Diaphragmatic surface:** composed mainly of posterior, inferior surfaces of right and left ventricles.
- External grooves (**sulci**) mark chamber borders: **coronary sulcus** separates atria from ventricles; **anterior interventricular sulcus** and **posterior interventricular sulcus** separate left and right ventricles; **interatrial groove** separates the two atria. Considerable adipose tissue usually found in epicardial connective tissue at these sulci (must be removed to expose grooves in fresh/preserved hearts). Sulci contain coronary arteries and veins.
- Atria vs. ventricles: atria receive venous blood continuing into ventricles; ventricles propel blood to peripheral tissues/lungs — functional differences linked to structural differences.
- Because the heart sits at an angle, the right atrium is anterior, inferior, and to the right of the left atrium. Left atrium curves posteriorly, forms most of the posterior heart surface superior to the coronary sulcus. Both atria have thin muscular walls.
- **Auricle:** when atria are not filled with blood, the anterior portion of each atrium deflates into a lumpy, wrinkled flap — named for resembling the external ear.
- **Clinical correlation — Myocarditis:** inflammation of the myocardium; commonly caused by viral infection, can affect young healthy people; also from autoimmune disease, environmental toxins, alcohol, certain medications, and chemotherapy/radiation (e.g., breast cancer therapy). If enough cardiac muscle cells damaged, can lead to chronic cardiomyopathy with impaired pumping power; blood clots may form in the heart, possibly leading to heart attack or stroke. Chronic cardiomyopathy may require a heart transplant.
- **Clinical correlation — Endocarditis:** inflammation of the endocardium; almost always bacterial infection (can be fungal); hearts with cardiac birth defects or damaged/abnormal valves are particularly susceptible. Begins when bacteria enter the bloodstream and settle on the endocardium — can follow dental surgery, hospital central venous line placement, or unsterile self-injection of drugs. Can destroy heart valves, requiring surgical replacement.

21.5 | Internal Anatomy and Organization of the Heart

- **Interatrial septum** separates the atria; **interventricular septum** separates the ventricles.

The Right Atrium

- Receives oxygen-poor (deoxygenated) venous blood from the systemic circuit via the **superior vena cava** and **inferior vena cava**.
- Superior vena cava opens into the posterior, superior right atrium; receives blood from head, neck, upper limbs, chest.
- Inferior vena cava opens into the posterior, inferior right atrium; receives blood from lower body.
- **Coronary veins** collect blood from the heart wall itself, deliver it to the **coronary sinus**, which opens into the posterior wall of the right atrium, inferior to the inferior vena cava opening.
- **Pectinate muscles** (comb-shaped) extend along the inner surface of the right auricle and across the anterior wall of the right atrium.
- **Foramen ovale**: an oval opening in the interatrial septum present from the 5th week of embryonic development until birth; allows blood to flow directly from right atrium to left atrium while lungs are developing. At birth, lungs expand/function, and the foramen ovale closes; permanently sealed within 3 months.
- **Clinical correlation — Patent foramen ovale**: if it remains open, blood passes from left atrium to right atrium and recirculates into the pulmonary circuit — reduces systemic circulation efficiency, elevates pulmonary vessel blood pressure. If uncorrected, leads to cardiac enlargement, fluid buildup in lungs, eventual heart failure.

The Right Ventricle

- Receives oxygen-poor venous blood from the right atrium through an opening guarded by three fibrous flaps (cusps) forming the **right atrioventricular (AV) valve**, aka **tricuspid valve**.
- Cusps attached on one side to the cardiac skeleton; free edges attached to **chordae tendineae** ("tendinous cords") arising from **papillary muscles** (cone-shaped muscular projections of the inner right ventricle surface). Chordae tendineae limit cusp movement when valve closes, preventing backflow into the right atrium.
- Internal surface contains irregular muscular ridges: **trabeculae carneae**.
- **Moderator band**: muscular ridge extending horizontally from the inferior interventricular septum, connecting to the anterior papillary muscle.
- Superior end of right ventricle tapers to the **conus arteriosus** (smooth-walled, cone-shaped pouch), ending at the **pulmonary valve** (pulmonary semilunar valve) — three thick semilunar (half-moon) cusps. Blood passes through this valve into the **pulmonary trunk** (first vessel of the pulmonary circuit). Pulmonary valve prevents backflow into the right ventricle when it relaxes.

The Left Atrium

- Receives oxygenated blood from pulmonary capillaries via small veins that unite to form four pulmonary veins (usually two per lung), emptying into the posterior left atrium.
- Differences from right atrium: (1) more cuboidal in shape; (2) left auricle is longer, narrower, more hook-shaped; (3) all pectinate muscles of the left atrium are contained within the left auricle.
- Blood leaves the left atrium via the **left atrioventricular (AV) valve**, aka **mitral valve** or **bicuspid valve** (two cusps, vs. three in the right AV valve) — permits oxygen-rich blood flow from left atrium into left ventricle, prevents reverse flow.

The Left Ventricle

- Wall is approximately **three times thicker** than the right ventricle wall — must generate enough pressure to push blood through the entire systemic circuit. Right ventricle only needs to push blood to the lungs and back, a total distance of **~30 cm (1 ft)**.
- Internal organization resembles the right ventricle, but: (1) trabeculae carneae are more prominent; (2) no moderator band; (3) since left AV valve has only two cusps, there are two large papillary muscles (rather than three).
- Blood leaves via the **aortic valve** (aortic semilunar valve) into the ascending aorta. Cusp arrangement similar to the pulmonary valve.

- **Aortic sinuses:** small saclike dilations at the base of the ascending aorta next to each aortic valve cusp; prevent cusps from sticking to the aortic wall when the valve opens. The right and left **coronary arteries** originate at the aortic sinuses.
- Aortic valve prevents backflow into the left ventricle once blood is pumped into the systemic circuit.
- From ascending aorta, blood flows into the aortic arch, then the descending aorta.
- **Ligamentum arteriosum:** fibrous band of connective tissue attaching the pulmonary trunk to the aortic arch; remnant of an important fetal vessel that once linked pulmonary and systemic circuits (ductus arteriosus).

Structural Differences between the Right and Left Ventricles

- Lungs are close to the heart; pulmonary vessels are short and wide, so the right ventricle doesn't need to work hard — its wall is relatively thin, resembling a pouch attached to the massive left ventricle wall.
- Right ventricle contraction moves toward the left ventricle wall, compressing blood within the right ventricle, forcing it through the pulmonary valve with minimal effort but relatively low pressure — appropriate since higher pressure would damage delicate pulmonary capillaries and force fluid into alveoli.
- Left ventricle must generate **4 to 6 times more force** to push blood through the systemic circuit. It has an extremely thick wall, round in cross section. On contraction: shortens/narrows, base-to-apex distance decreases, chamber diameter decreases (like squeezing/rolling a toothpaste tube) — generates enough pressure to force open the aortic valve.
- As the left ventricle contracts, it bulges into the right ventricular cavity, improving right ventricle pumping efficiency — individuals with severe right ventricle damage may survive because left ventricle contraction helps push blood into the pulmonary circuit.

The Structure and Function of Heart Valves

- Valves are folds of endocardium extending into the openings between atria/ventricles; open/close to prevent backflow, maintaining one-way blood flow.
- Each AV valve has four components: (1) ring of connective tissue attached to cardiac skeleton, (2) connective tissue cusps closing the chamber opening, (3) chordae tendineae attaching cusp margins to papillary muscles, (4) papillary muscles that tense the chordae tendineae. Tension in papillary muscles/chordae tendineae prevents cusps from swinging too far open into the atria — prevents regurgitation (backflow) during ventricular contraction.
- **Pulmonary valve** (pulmonary semilunar valve): junction between right ventricle and pulmonary artery. **Aortic valve** (aortic semilunar valve): junction between left ventricle and ascending aorta. These lack chordae tendineae because three symmetrical cusps support one another like tripod legs.
- **Clinical correlation — Heart murmur:** soft whooshing/swishing sound from turbulent blood flow in/near the heart during a heartbeat. Most murmurs are "innocent" (no clinical significance) — e.g., pregnant women often develop innocent murmurs due to increased blood volume/cardiac output while valve size stays the same. Congenital cardiac defects (atrial septal defects, ventricular septal defects — holes in the interatrial/interventricular septum) cause abnormal flow direction during systole, generating a murmur.
- **Clinical correlation — Stenotic/incompetent valves:** **Valvular stenosis** = narrowing of the valve opening, reduces blood flow. **Valvular insufficiency** = regurgitation of blood from incomplete valve closure.
- Health professionals use a **stethoscope** to listen to heart sounds; sounds may be muffled by the pericardium, surrounding tissues, chest wall, so stethoscope placement doesn't always correspond exactly to valve position.

21.6 | Coronary Blood Vessels

- Heart works continuously; cardiac muscle cells require reliable oxygen/nutrient supplies. **Coronary circulation** supplies blood to heart muscle tissue.
- During maximum exertion, blood flow to the heart may increase to **nine times resting levels**.
- Left and right coronary arteries originate at the base of the ascending aorta, within the aortic sinus — first branches from the aorta. Blood pressure here is the highest in the systemic circuit, guaranteeing continuous flow to active cardiac muscle.

Right Coronary Artery (RCA)

- Branches off the ascending aorta, turns right, lies within the coronary sulcus, passes between the right auricle and the pulmonary trunk.
- **60%** of the time, the RCA is the "dominant" coronary artery (gives off the posterior interventricular branch).
- Typically supplies: (1) right atrium, (2) portion of left atrium, (3) interatrial septum, (4) entire right ventricle, (5) variable portion of left ventricle, (6) postero-inferior 1/3 of interventricular septum, (7) portions of the conducting system (SA node).
- Gives off atrial branches supplying right atrium and part of left atrium.
- **Right marginal branch:** forms near the right heart border, extends toward the apex along the anterior right ventricle surface; supplies right atrium, interatrial septum, right ventricle.
- Continuing across the posterior surface, RCA gives off the **posterior interventricular branch** (posterior descending artery), continuing toward the apex within the posterior interventricular sulcus; supplies the interventricular septum and adjacent ventricular portions.
- A small branch near the RCA base penetrates the atrial wall to reach the **SA node** (cardiac pacemaker). A small branch to the **AV node** also originates from the RCA near the posterior interventricular branch.

Left Coronary Artery (LCA)

- Lumen usually larger in diameter than the RCA.
- Supplies: (1) most of the left ventricle, (2) small segment of the right ventricle, (3) most of the left atrium, (4) anterior 2/3 of the interventricular septum.
- Forms the **anterior interventricular branch** (left anterior descending) — supplies ventricular myocardium and anterior 2/3 of interventricular septum; small branches often continuous with the posterior interventricular branch of the RCA.
- **Circumflex branch:** curves left within the coronary sulcus toward the posterior heart surface; gives rise to diagonal branches supplying portions of the left ventricle.
- LCA typically forms the **left marginal branch**, reaching the apex, supplying much of the left ventricle.
- RCA (upon reaching the posterior surface) forms the **posterior left ventricular branch** — small, variable, sometimes absent; if absent, often replaced by the posterior interventricular branch of the circumflex artery. Distal circumflex branch portions often fuse with small RCA branches.
- **Anastomoses:** interconnections between coronary arteries; keep ventricular muscle blood supply relatively constant despite pressure fluctuations in the left/right coronary arteries.

Coronary Veins

- **Great cardiac vein** and **middle cardiac vein** collect blood from smaller veins draining myocardial capillaries, deliver it to the **coronary sinus** — a large, thin-walled vein in the posterior coronary sulcus. Coronary sinus drains into the right atrium, inferior to the inferior vena cava opening.
- Other cardiac veins emptying into the great cardiac vein or coronary sinus:
 - **Posterior vein of the left ventricle:** drains area served by the circumflex branch of the LCA.
 - **Middle cardiac vein:** drains area supplied by the posterior interventricular branch of the LCA (as stated in text).
 - **Small cardiac vein:** receives blood from posterior surfaces of right atrium and ventricle.
- **Anterior cardiac veins:** drain the anterior right ventricle surface; empty directly into the right atrium, bypassing the coronary sinus.
- **Clinical correlation — Coronary Artery Disease (CAD):** leading cause of death for men and women in the US. Occurs when cardiac arteries harden/narrow from cholesterol, calcium, plaque buildup (**atherosclerosis**), decreasing blood supply to heart muscle, causing **cardiac ischemia**. Ischemia often causes **angina** (crushing, smothering chest discomfort, may radiate to back, neck, jaw, left arm).
 - If a blood clot forms at a plaque site and suddenly blocks a vessel (coronary thrombosis), this affects the heart's ability to contract properly, impairs the conduction system, can lead to arrhythmias and heart failure.
 - Treatment: restoring blood flow with a thrombolytic (clot dissolver) or coronary angioplasty (surgical repair). Preventable causes of CAD: smoking, obesity, hypertension, hypercholesterolemia, diabetes, sedentary lifestyle.

- **Percutaneous balloon angioplasty**: threading a catheter past atherosclerosis, inflating a balloon to compress plaque against the vessel wall, then inserting a metal stent to keep the area open.
- **Coronary artery bypass graft (CABG)**: surgical replacement of damaged coronary artery sections with venous grafts, typically using the great saphenous vein of the leg or the internal thoracic artery of the chest wall.
- **Electrocardiogram (ECG)**: can demonstrate cardiac ischemia.

21.7 | The Coordination of Cardiac Contractions

- Function of any pump: (1) develop pressure, (2) move a fluid volume in a specific direction at an acceptable speed.
- Heart works in cycles of contraction (**systole**) and relaxation (**diastole**); pressure within each chamber rises/falls within each cycle.
- AV, aortic, and pulmonary valves ensure one-way blood flow: blood flows out of an atrium only when the AV valve is open and atrial pressure > ventricular pressure; blood flows from a ventricle into the aorta/pulmonary trunk only when that valve is open and ventricular pressure > arterial pressure.
- Proper heart function depends on proper timing of atrial/ventricular contractions, provided by the pacemaking and conduction systems.
- Cardiac muscle contracts on its own without neural/hormonal stimulation (even a heart removed for transplant continues beating unless kept chilled in preservation solution). This ability = **autorhythmicity** (or automaticity).
- Cardiac contractions coordinated by specialized conducting cells — two distinct populations: (1) **nodal cells** establish the rate of cardiac contraction, (2) **conducting cells** distribute the contractile stimulus to the myocardium.

Conducting System Components

- **Sinoatrial (SA) node**: contains pacemaker cells that initiate the electrical impulse resulting in a heartbeat.
- **Internodal pathways**: conducting fibers in the atrial wall that conduct the impulse to the AV node while simultaneously stimulating atrial contraction.
- **Atrioventricular (AV) node**: slows the electrical impulse arriving from the internodal pathways.
- **AV bundle**: conducts the impulse from the AV node to the bundle branches.
- **Left bundle branch**: extends toward the apex, radiates across the inner surface of the left ventricle.
- **Right bundle branch**: extends toward the apex, radiates across the inner surface of the right ventricle.
- **Moderator band**: relays the stimulus through the ventricle to the papillary muscles, tensing the chordae tendineae before the next cardiac cycle.
- **Purkinje fibers**: convey impulses very rapidly to the contractile cells.
- Sequence of impulse conduction with approximate timing:
 - Time = 0: SA node depolarizes, atrial activation begins.
 - ~50 msec: depolarization spreads across atrial surfaces, reaches AV node.
 - ~150 msec: AV node delays spread of electrical activity to the AV bundle by 100 msec; atrial contraction begins.
 - ~175 msec: impulses travel along the AV bundle within the interventricular septum to the apex; impulses also spread to papillary muscles of the right ventricle via the moderator band; atrial contraction completes, ventricular contraction begins.
 - ~225 msec: impulse distributed by Purkinje fibers (subendocardial branches), relayed throughout the ventricular myocardium.

21.8 | The Cardiac Cycle

- The period between the start of one heartbeat and the beginning of the next = a single **cardiac cycle**; includes alternating contraction/relaxation periods. For any one chamber, the cycle has two phases:

- **Systole** (contraction): chamber ejects blood into an adjacent chamber or an arterial trunk.
- **Diastole** (relaxation): chamber fills, is relaxed.
- Cardiac cycle phase overview:
 - Atrial systole ends; atrial diastole begins (continues until start of next cardiac cycle).
 - Ventricular diastole—late: pressure builds to open the semilunar valves.
 - Ventricular systole—second phase: as ventricular pressure rises and exceeds arterial pressure, the semilunar valves open and blood is ejected.
 - Ventricular diastole—early: as ventricles relax, ventricular blood pressure drops until reverse blood flow pushes the semilunar valve cusps together; blood then flows into the relaxed atria.
- Plasma membranes of nodal cells spontaneously depolarize to threshold; nodal cells possess intercellular junctions electrically coupling them to each other, conducting fibers, and cardiac muscle cells. Nodal cell depolarization generates an action potential that travels through the conducting system, reaching all cardiac muscle tissue, causing contraction (a full heartbeat/cardiac cycle).
- Not all nodal cells depolarize at the same rate — normal contraction rate is determined by which nodal cells reach threshold first; these rapidly depolarizing cells = **pacemaker cells**, found in the **SA node** (cardiac pacemaker).
- SA node located in the posterior wall of the right atrium, near the entrance of the superior vena cava.
- Pacemaker cells depolarize spontaneously and rapidly, generating **80–100 action potentials per minute** — this is the resting/inherent heart rate before autonomic modulation.
- Factors that change the SA node's resting potential or spontaneous depolarization rate alter heart rate; ANS activity affects nodal cell activity.
 - Parasympathetic neurons release **ACh** → slows spontaneous depolarization → decreases heart rate.
 - Sympathetic neurons release **NE** → increases depolarization rate → increases heart rate.
- Under normal resting conditions, parasympathetic activity reduces heart rate from the inherent nodal rate of 80–100 impulses/min to a more leisurely **70–80 beats/min**.
- **Clinical correlation — Bradycardia**: slower-than-normal heart rate. **Tachycardia**: faster-than-normal heart rate. Clinical definitions vary with the individual's normal resting rate and conditioning.
- SA node cells electrically connected to the larger **AV node** through conducting fibers in the atrial walls.

21.9 | Autonomic Control of Heart Rate

- Sympathetic and parasympathetic ANS divisions innervate the heart through the **cardiac plexus**. Both divisions innervate the SA and AV nodes, plus atrial/ventricular cardiac muscle cells and smooth muscle in cardiac blood vessel walls.
- Effects on nodal tissues:
 - **NE release**: increases heart rate and contraction force by stimulating beta receptors on nodal cells and contractile cells.
 - **ACh release**: decreases heart rate and contraction force by stimulating muscarinic receptors on nodal cells and contractile cells.
- **Cardiac centers** of the medulla oblongata contain the ANS centers for cardiac control:
 - **Cardioacceleratory center**: activates sympathetic neurons.
 - **Cardioinhibitory center**: activates parasympathetic neurons.
- Cardiac centers receive inputs from higher centers, especially the parasympathetic/sympathetic headquarters in the hypothalamus.
- Cardiac centers modify heart rate/cardiac output through the **vagus nerves** (parasympathetic) and **cardiac nerves** (sympathetic).
 - Parasympathetic pathway: nucleus of the vagus nerve → vagus nerve (X) → cardioinhibitory center.
 - Sympathetic pathway: spinal cord → sympathetic chain (cervical ganglia and postganglionic ganglia T1–T4) → cardiac nerve → cardioacceleratory center.

- Sensory info about cardiovascular status arrives at cardiac centers via visceral sensory fibers from the **glossopharyngeal (IX)** and **vagus (X)** nerves — afferent fibers originate in baroreceptors (sensitive to blood pressure) and chemoreceptors (sensitive to dissolved gas concentrations).
- Cardiac centers respond quickly to adjust cardiac performance, maintaining adequate circulation to vital organs (e.g., brain). E.g., a drop in blood pressure or rise in CO₂ indicates the heart must work harder — cardiac centers respond by increasing heart rate and contraction force via sympathetic activation.
- Mnemonic: "SSS" — Sympathetic input Speeds and Strengthens the heartbeat.

Clinical Case Wrap-Up & Related Clinical Terms (Ch 21)

- Ellen's ECG showed changes resembling a myocardial infarction; cardiac catheterization revealed her left ventricle was abnormally large and balloon-shaped with apical wall hypokinesia (decreased motion during systole), causing decreased cardiac output; coronary vessels appeared normal.
- Diagnosis: **takotsubo cardiomyopathy** ("broken heart syndrome") — named for the ventricle's end-systole shape resembling a Japanese fisherman's octopus pot (tako-tsubo); associated with severe emotional stress; caused by the sympathetic nervous system releasing norepinephrine in "overdrive," affecting the heart. Recovery typically occurs with supportive care alone.
- Related clinical terms:
 - **Artificial pacemaker:** small battery-operated device keeping the heart beating in regular rhythm; may be permanently implanted or external/temporary.
 - **Asystole:** absence of cardiac activity — no contraction, no output.
 - **Cardiomyoplasty:** surgical procedure using stimulated latissimus dorsi wrapped around the ventricles to assist cardiac function via an implanted burst-stimulator, contracting during systole.
 - **Cardiomegaly:** enlarged heart; a sign of another underlying condition (stress, weakening of heart muscle, coronary artery disease, abnormal heart rhythms).
 - **Fibrillation:** fast twitching of heart muscle fibers with little/no coordinated movement of the muscle as a whole. Atrial fibrillation occurs in the atria, with both atria and ventricles beating out of sync.
 - **Heart block:** delay in the normal electrical pulses that cause the heart to beat.
 - **Palpitation:** irregular and rapid beating of the heart.
 - **Sick sinus syndrome:** group of heart rhythm disorders where the SA node does not work properly to regulate heart rhythm.
 - (Also referenced: implantable defibrillator that detects a shockable rhythm and delivers a shock to restore normal rhythm; balloon-tipped catheter used to enlarge a narrowed artery; shortness of breath from inability of the heart to maintain adequate circulation.)

Ch 22 — Vessels and Circulation, Great Vessels (p. 567–585)

Introduction

- Cardiovascular system = closed system circulating blood throughout the body; two vessel groups: **pulmonary circuit** (supplies lungs) and **systemic circuit** (supplies rest of body). Heart pumps into both simultaneously.
- Pulmonary circuit begins at the pulmonary valve, ends at the entrance to the left atrium.
- Systemic circuit begins at the aortic valve, ends at the entrance to the right atrium.
- After vessels enter an organ, further branching creates several hundred million tiny arteries supplying more than **10 billion capillaries**, each about the diameter of a single red blood cell.
- Combined length of all capillaries in the body, placed end to end, is estimated at **5000 to 25,000 miles** — could cross the continental US or circle the globe.
- All chemical/gaseous exchange between blood and interstitial fluid occurs across capillary walls.

22.1 | Histological Organization of Blood Vessels

- Walls of arteries and veins contain three distinct layers, superficial to deep: (1) outer **adventitia** (tunica externa), (2) middle **media** (tunica media), (3) inner **intima** (tunica intima).

1. **Adventitia**: thick connective tissue sheath, chiefly collagen fibers with scattered elastic fiber bands. In veins, this layer is usually thicker than the media. Blends into adjacent tissues, stabilizing/anchoring the vessel.

2. **Media**: middle layer; concentric layers of smooth muscle supported by loose connective tissue framework. Smooth muscle encircles the vessel lumen. Sympathetic ANS stimulation → contraction → **vasoconstriction** (reduces luminal diameter). Relaxation → **vasodilation** (increases diameter). Smooth muscle also responds to local stimuli (pH, pCO₂, pO₂ changes). Collagen fibers bind media to intima and adventitia. Arteries have a thin **external elastic membrane** between media and adventitia.

3. **Intima**: innermost layer; endothelial lining + underlying connective tissue layer with variable elastic fibers. In arteries, the outer margin of the intima contains a thick elastic fiber layer = **internal elastic membrane**; largest arteries have thicker intima.

- Vessel walls are too thick for diffusion between bloodstream and surrounding tissue; large vessel walls contain their own small arteries/veins (**vasa vasorum**, "vessels of vessels") supplying the smooth muscle cells, fibroblasts, fibrocytes of the media/adventitia.

Distinguishing Arteries from Veins

- Arteries and veins in the same body region typically lie side by side within a shared connective tissue sheath.
- **Vessel walls**: artery walls are thicker than corresponding vein walls; artery media has more smooth muscle and elastic fibers (resists cardiac pumping pressure).
- **Vessel lumen**: artery walls contract, constricting the lumen — arterial lumen appears smaller than accompanying vein in section; arteries retain circular shape in histological section (endothelium folds when constricted, giving a "pleated" appearance); veins collapse when cut, appearing flattened/distorted, lacking folds.
- **Valves**: veins typically contain valves (prevent backflow toward capillaries); arteries lack valves.

Arteries — types (elastic → muscular → arterioles)

- **Elastic arteries**: large vessels, luminal diameter up to ~2.5 cm; transport large blood volumes away from the heart. Examples: pulmonary artery, aorta and major branches (brachiocephalic trunk, carotid, subclavian, common iliac arteries). Intima relatively thick (endothelial lining + basal lamina of collagen/elastic fibers); internal elastic membrane separates intima from media. Media has high density of elastic fibers, relatively few smooth muscle cells — extremely resilient, tolerates cardiac cycle pressure changes; smooth muscle in elastic artery media does NOT contract in response to sympathetic/local stimulation. During ventricular systole, elastic arteries stretch as pressure rises; during diastole, elastic recoil slows the pressure drop and forces blood onward toward capillaries.
 - Aorta/brachiocephalic artery walls: not very thick relative to diameter but extremely resilient. Average luminal diameter ~1.5 cm, wall thickness ~1 mm (per spotlight figure).
- **Muscular arteries** (medium-sized arteries): transport blood to skeletal muscles and internal organs. Thicker media with greater % smooth muscle than elastic arteries; prominent internal elastic membrane separates intima from media. Sympathetic ANS + local stimulation control luminal diameter, allowing independent regulation of blood flow to each organ. Example: radial/ulnar arteries — average luminal diameter ~4 mm, wall thickness ~1 mm.
- **Arterioles**: considerably smaller than muscular arteries; average luminal diameter ~30 µm, wall thickness ~6 µm; visible only with a microscope. Thin adventitia; media = one or two smooth muscle layers that may not form a complete layer. Arterioles (and smaller muscular arteries) change luminal diameter in response to local conditions or sympathetic/endocrine stimulation; arterioles control blood flow between arteries and capillaries.
- Elastic and muscular arteries interconnect seamlessly; vessel characteristics change gradually with distance from the heart (largest muscular arteries have considerable elastic tissue; smallest resemble heavily muscled arterioles).

- **Clinical correlation — Arteriosclerosis:** hardening of the arteries; several types. **Atherosclerosis** = most common type; deposition of fatty plaques on inner arterial walls, beginning with endothelial damage; develops gradually, likely starting early in life; can affect vessels anywhere in the body.
 - Coronary vessels involved → coronary artery disease (CAD) → angina, myocardial infarction.
 - Carotid/cerebral arteries → transient ischemic attack (TIA, "mini stroke," temporary blockage) or stroke (permanent brain damage) if it progresses.
 - Extremity arteries (esp. lower legs) → peripheral artery disease (PAD): intermittent claudication (limping), ischemia, calf muscle pain from walking; if blood supply lost completely, gangrene can result.
 - Renal arteries → high blood pressure, potentially kidney failure.
 - Non-preventable risk factors: aging, family history, certain inflammatory diseases. Preventable risk factors: smoking (worst/most preventable), high blood pressure, diabetes, obesity, sedentary lifestyle, high cholesterol/triglycerides.

Capillaries

- Smallest, most delicate blood vessels; only vessels whose walls allow nutrient/waste exchange between blood and interstitial fluid; thin walls = short diffusion distances = fast exchange.
- Average luminal diameter ~8 μm , close to a single red blood cell.
- Blood flows slowly through capillaries, allowing time for diffusion/active transport across walls. Substances cross by diffusing across endothelial cells or passing through gaps between adjacent endothelial cells.
- **Continuous capillaries:** found in all tissues except epithelia and cartilage; complete endothelial lining, cells connected by tight junctions and desmosomes.
- **Fenestrated capillaries:** contain "windows"/pores from incomplete/perforated endothelial lining; "Swiss cheese" appearance; pores allow peptides and small proteins to pass; permit rapid fluid/solute exchange. Examples: choroid plexus of the brain, capillaries in endocrine glands, gallbladder, kidney, pancreas, some intestinal locations.
- **Sinusoids** (discontinuous capillaries): flattened, irregularly shaped, fenestrated capillaries with gaps between adjacent endothelial cells; basal lamina thinner or absent (absent entirely in the liver). Blood moves slowly through sinusoids, maximizing absorption/secretion time. Found in liver, bone marrow, spleen, adrenal glands.
- Four mechanisms of material exchange across capillary walls: (1) diffusion across endothelial cells (lipid-soluble materials, gases, water via osmosis); (2) diffusion through gaps between endothelial cells (water, small solutes; larger solutes in sinusoids); (3) diffusion through pores in fenestrated capillaries/sinusoids (water, solutes); (4) vesicular transport by endothelial cells (endocytosis at luminal side, exocytosis at basal side).
- **Capillary bed:** collection of capillaries; single arteriole gives rise to dozens of capillaries emptying into several venules. Smooth muscle cells within arterioles cycle contracting/relaxing (~dozen times/min), so blood flow occurs in pulses; flow route within a bed changes moment to moment — controlled at the tissue level, called **autoregulation**.
- **Arterial anastomosis:** more than one muscular artery (**collaterals**) supplies a capillary bed, fusing together rather than forming a series of arterioles; found in brain, heart, stomach, other high-demand organs/regions; guarantees reliable blood supply if one artery is blocked.
- **Arteriovenous anastomoses:** direct connections between arterioles and venules; common in visceral organs and joints where body position changes hinder flow; smooth muscle in walls regulates blood reaching the capillary bed — when dilated, blood bypasses the capillary bed, flowing directly into venous circulation.
- **Thoroughfare channels:** passageways in some connective tissues connecting arterioles with venules, lined with smooth muscle that constricts/dilates to regulate flow.
- **Metarterioles** (precapillary sphincters): guard capillary entrances in mesenteries; represent thoroughfare channels within mesenteric circulation (existence outside mesenteries not well supported in literature).

Veins

- Collect blood from all tissues/organs, return it to the heart. Walls thinner than corresponding arteries (lower venous blood pressure). Classified by size. Despite thinner walls, veins generally have larger luminal diameters than corresponding arteries.

- **Venules:** smallest veins, collect blood from capillaries. Smallest ("postcapillary venules") resemble expanded capillaries; venules <50 µm total diameter lack a complete media (only isolated smooth muscle cells deep to the intima). Venules >50 µm diameter have media with one or two thin smooth muscle layers dominated by connective tissue.
- **Medium-sized veins:** luminal diameter 2–10 mm; most deep veins (radial, tibial, popliteal) fall in this category; typically found in the same connective tissue sheath alongside a muscular artery. Three-layer wall organization most visible here: intima = endothelium on basal lamina; media = several smooth muscle layers + collagen/elastic connective tissue; adventitia typically thicker than media, elastic/collagenous fibers. **Valves** are a characteristic feature of medium-sized veins.
- **Venous valves:** form from infoldings of the intima; prevent backflow (like heart valves); compartmentalize blood, dividing the weight of the blood column between compartments. As long as functioning normally, skeletal muscle movement squeezes blood toward the heart — the **skeletal muscle pump**.
- **Large veins:** include the superior and inferior venae cavae and tributaries (subclavian, renal, mesenteric, portal veins) within the abdominopelvic/thoracic cavities. Intima and media are small, difficult to distinguish; adventitia is the thickest layer, containing collagen/elastic fibers and isolated longitudinal smooth muscle cells. Large veins do NOT have valves; changes in body-cavity pressure help move blood toward the heart. (Spotlight figure: SVC/IVC = "great veins," average luminal diameter ~2 cm, wall thickness ~2 mm.)
- **Venoconstriction:** sympathetic nerve stimulation of smooth muscle in medium-sized vein walls (systemic circuit) during serious hemorrhage; reduces venous blood volume, increases arterial/capillary volume, maintaining near-normal arterial blood volume despite blood loss — sustains flow to active skeletal muscles and delicate organs (e.g., brain).

22.2 | The Distribution of Blood

- Blood volume unevenly distributed: heart + arteries + capillaries contain ~30–35% of blood volume (~1.5 L); venous system contains the rest (~65–70%, ~3.5 L).
- Veins are much more distensible than arteries (thinner walls, more elastic tissue, less smooth muscle) — for a given blood pressure rise, a typical vein stretches ~8 times as much as a corresponding artery.
- **Capacitance** = relationship between blood volume a vessel contains and blood pressure; veins = **capacitance vessels**, acting as blood reservoirs accommodating large blood volume changes.
- Pulmonary circuit contains ~9% of total blood volume at any moment. Systemic circuit contains ~84% of total blood volume.
- Pulmonary circuit: base of pulmonary trunk to lungs is only ~15 cm (6 in.) apart — short compared to systemic circuit.

22.3 | Functional Patterns of Blood Vessel Distribution

- Three important functional patterns:
1. Peripheral distribution of arteries/veins on left/right sides is usually identical, except near the heart where the largest vessels connect to atria/ventricles.
 2. A single vessel may have several different names as it crosses specific anatomical boundaries (enables precise anatomical description as a vessel extends into the periphery).
 3. Anastomoses are often found between arteries and veins — these reduce the impact of temporary or permanent occlusion of a single vessel.

22.4 | The Pulmonary Circuit

- Blood entering the right atrium returns from peripheral tissue/myocardial capillary beds (O₂ released, CO₂ absorbed there). Passes through right atrium → right ventricle → pulmonary valve → pulmonary trunk (start of pulmonary circuit).
- Pulmonary circuit begins at the pulmonary valve, ends at entrance to the left atrium; in this circuit oxygen is replenished, CO₂ released.

- Pulmonary circuit arteries carry **deoxygenated** blood (unlike systemic arteries) — color-coded diagrams typically show pulmonary arteries in blue (same as systemic veins).
- Pulmonary trunk curves over the superior heart border, dividing into left and right **pulmonary arteries**. These enter the lungs, branch repeatedly into smaller arteries, then **pulmonary arterioles**, supplying capillary networks around alveoli. Alveolar walls are thin enough for gas exchange between capillary blood and inspired air.
- Oxygenated blood leaves alveolar capillaries, enters venules merging into larger vessels toward the **pulmonary veins** — four total, typically two from each lung — emptying into the left atrium, completing the pulmonary circuit.

22.5 | The Systemic Circuit — Major Arteries (Aorta and Great Vessel Branches)

- Systemic circuit begins at the aortic valve, ends at the entrance to the right atrium; supplies capillary beds throughout the body not supplied by the pulmonary circuit.
- Mnemonic for aortic arch branches: "ABCs" — **A**ortic arch, **B**rachiocephalic trunk, **C**ommon carotid artery, **S**ubclavian artery.

The Ascending Aorta and the Aortic Arch

- The **aorta** begins at the aortic valve of the left ventricle. First segment = **ascending aorta**. Left and right coronary arteries originate at the base of the ascending aorta, just superior to the aortic valve.
- **Aortic arch** (arch of the aorta): curves like a cane handle, passing superiorly/posteriorly across the heart's superior surface; connects ascending aorta to descending aorta.
- Three elastic arteries originate along the aortic arch, delivering blood to the head, neck, shoulders, upper limbs:
 1. **Brachiocephalic trunk**: only one (unpaired); ascends a short distance, then (posterior to the right sternoclavicular joint) divides into the **right subclavian artery** and **right common carotid artery**.
 2. **Left common carotid artery**: arises separately from the aortic arch.
 3. **Left subclavian artery**: arises separately from the aortic arch.
- The vessels on the left side are mirror equivalents of the right-side branches (right subclavian/right common carotid derive from the brachiocephalic trunk; left subclavian/left common carotid arise directly).

The Subclavian Arteries

- Arch over the apex (superior tip) of the right/left lungs; supply the upper limbs, chest wall, shoulders, back, brain, spinal cord.
- Three major branches arise before the subclavian arteries leave the thoracic cavity:
 1. **Thyrocervical trunk**: supplies muscles/tissues of the neck, shoulder, upper back.
 2. **Internal thoracic artery**: supplies the pericardium and anterior chest wall.
 3. **Vertebral artery**: supplies the brain and spinal cord.
- After leaving the thoracic cavity, passing over the first rib, each subclavian artery becomes the **axillary artery** (supplies pectoral region/axilla muscles), then the **brachial artery** (supplies the upper limb) — [detailed forearm/hand branches such as deep brachial, ulnar collateral/recurrent, radial/ulnar arteries, palmar arches are part of upper-limb anatomy, beyond the thoracic "great vessels" scope].

The Carotid Arteries and Blood Supply to the Brain

- Common carotid arteries ascend deep in the neck, lateral to the trachea. At the level of the larynx, each divides into an **external carotid artery** and **internal carotid artery**.
- **Carotid sinus**: located at the base of the internal carotid artery (often extends along part of the common carotid); contains baroreceptors and chemoreceptors involved in cardiovascular regulation.
- **External carotid arteries**: supply the neck, pharynx, esophagus, larynx, lower jaw, face.
- **Internal carotid arteries**: enter the skull through the carotid canals of the temporal bones, delivering blood to the brain; each ascends to the level of the optic nerves, dividing into three branches: (1) ophthalmic artery (eye),

(2) anterior cerebral artery (frontal/parietal lobes), (3) middle cerebral artery (midbrain, lateral cerebral hemisphere surfaces).

- The brain is very sensitive to circulatory interruption: unconsciousness after several seconds; permanent neural damage possible after **4 minutes**. Brain receives blood from two sources — vertebral arteries and internal carotid arteries — reducing interruption risk.
- **Vertebral arteries:** branch from the subclavian arteries, ascend within the transverse foramina of cervical vertebrae, enter the cranium at the foramen magnum; the two vertebral arteries fuse along the ventral medulla oblongata surface to form the **basilar artery**, which continues along the ventral pons surface, branches repeatedly, then divides into the **posterior cerebral arteries** → **posterior communicating arteries**.
- Internal carotid arteries branch to supply the anterior half of the cerebrum; the rest of the brain is supplied by the vertebral arteries. Internal carotid arteries and the basilar artery interconnect, forming a ring-shaped anastomosis: the **cerebral arterial circle** (circle of Willis), encircling the infundibulum of the pituitary gland — allows the brain to receive blood from carotids and/or vertebrals, reducing interrupted-circulation likelihood.
- **Clinical correlation — Aneurysm:** bulge/weakness in an artery wall; can occur anywhere, including brain and heart (focus here: aorta). Aortic wall is very elastic, stretching/recoiling with blood flow. Aortic aneurysms classified by location: thoracic aortic aneurysm (ascending or descending) or abdominal aortic aneurysm.
 - Many abdominal aortic aneurysms cause no early symptoms, discovered incidentally on x-ray/ultrasound; some can be palpated or auscultated in thin individuals.
 - Screening recommendation: one-time ultrasound for people over 65 with family history, and men over 65 who ever smoked.
 - If ruptured, prognosis is very poor — mortality rate over **90%**. Preventive surgical treatment recommended for aneurysms **>5 cm** diameter that are enlarging.
 - Open surgical repair: large midline incision, aorta cross-clamped above/below the aneurysm to stop flow, aneurysm sac opened, synthetic tube graft inserted and sutured, aorta wrapped around the graft and closed.
 - **Endovascular repair** (less invasive): small incision into the femoral artery, guide wire passed to the aneurysm site, sheath slid over the wire, stent graft inserted through the femoral artery and advanced up the aorta, expanded and attached to the internal aortic wall to prevent further expansion.

The Descending Aorta

- Continuous with the aortic arch. The diaphragm divides the descending aorta into a superior **thoracic aorta** and inferior **abdominal aorta**.
- **Thoracic aorta:** begins at the level of vertebra T1 (per text; note some editions cite T5), lies slightly left of midline, gradually shifts toward midline, exits the posterior mediastinum, passes through the diaphragm at the level of vertebra **T12**. Supplies blood to thoracic organs (except the heart), chest/diaphragm muscles, and the thoracic portion of the spinal cord.
- Branches of the thoracic aorta grouped anatomically as **visceral** or **parietal**:
 - **Visceral branches** (supply chest organs): **bronchial arteries** (conducting passageways of the lungs), **pericardial arteries** (pericardium), **mediastinal arteries** (general mediastinal structures), **esophageal arteries** (esophagus).
 - **Parietal branches** (supply the chest wall): **intercostal arteries** (chest muscles, vertebral column area), and the superior [phrenic artery — text cuts off, likely "superior phrenic arteries" supplying the diaphragm].

The Abdominal Aorta (brief, boundary content)

- Begins immediately inferior to the diaphragm, descends slightly left of the vertebral column, posterior to the peritoneal cavity; often surrounded by an adipose tissue cushion.
- At the level of vertebra L4, the abdominal aorta splits into the right and left **common iliac arteries**, supplying deep pelvic structures and the lower limbs. This splitting region = the **terminal segment** of the aorta. [Further abdominal aorta branch detail belongs to the Abdomen unit.]

Ch 23 — Thoracic Duct / Cisterna Chyli (p. 603–609)

23.1 | An Overview of the Lymphatic System

- **Lymphatic system** (lymphoid system): functionally part of the circulatory and immune systems; monitors and transports **lymph** (fluid connective tissue).
- Vessels carrying lymph = **lymphatic vessels**; cells suspended within lymph (and blood) = **lymphocytes**.
- Lymph consists of: (1) interstitial fluid resembling blood plasma but with lower protein concentration, (2) lymphocytes (immune response cells), (3) macrophages (phagocytic cells).
- Lymphatic vessels often begin within or pass through lymphatic tissues/organs (contain large numbers of lymphocytes, macrophages, and often lymphatic stem cells).
- **Functions of the lymphatic system:**

1. **Produces, maintains, and distributes lymphocytes:** produced/stored in lymphatic organs (spleen, thymus, red bone marrow). Lymphatic tissues/organs classified as:

- **Primary lymphatic structures:** sites where lymphocytes are produced and mature — red bone marrow and the adult thymus.
- **Secondary lymphatic structures:** sites where lymphocytes are activated to produce additional lymphocytes of the same type (e.g., activated B cells dividing to fight infection); located at "the front lines" where pathogens are first encountered — examples: tonsils, lymph nodes.

2. **Maintains normal blood volume and eliminates local variations in interstitial fluid chemical composition:** blood pressure at the proximal end of a systemic capillary $\approx 35 \text{ mm Hg}$, forcing water/solutes out of plasma into interstitial fluid. Approximately **3.6 L (72% of total blood volume)** enters interstitial fluid each day; lymphatic vessels return an equal volume back to the bloodstream, so this movement normally goes unnoticed. This circulation eliminates regional interstitial fluid composition differences. Because so much fluid moves through daily, a break in a major lymphatic vessel can cause a rapid, potentially fatal decrease in blood volume.

3. **Provides an alternate route for transporting hormones, nutrients, and wastes:** e.g., lipids absorbed by the digestive tract are carried to the bloodstream by lymphatic vessels rather than across capillary walls.

23.2 | Structure of Lymphatic Vessels

- Cardiovascular blood vessels form a complete circuit (begin/end at the heart); lymphatic vessels (lymphatics) carry lymph only from peripheral tissues to the venous system (one-way).
- Lymphatic vessels range in size from small-diameter lymphatic capillaries to large-diameter collecting vessels.

Lymphatic Capillaries

- The lymphatic network begins with closed tubes called **lymphatic capillaries**. Differ from vascular (blood) capillaries:

1. Larger luminal diameters.
2. Thinner walls; endothelial cells lack a continuous basal lamina.
3. Flat or irregular outline.
4. Endothelial cells attached to surrounding connective tissue by fine filaments (anchored to an incomplete basal lamina) — these filaments keep passageways open when interstitial pressure increases.
5. Greater permeability because endothelial cells overlap rather than being tightly bound together.
 - The overlapping endothelial cell region acts as a one-way valve: permits fluids, solutes (including proteins), viruses, bacteria, and cell debris to enter, but prevents their return to intercellular spaces.
 - Lymphatic capillaries present in most tissues; especially numerous in connective tissue deep to skin/mucous membranes and in digestive tract mucosa/submucosa. Absent in areas without a blood supply (e.g., cornea of the eye).
 - **Lacteals:** lymphatic capillaries in the small intestine that transport lipids absorbed by the digestive tract.

- In living tissue, characteristic colors: arteries bright red, veins dark red, lymphatics a pale golden color.
- **Clinical correlation — Lymphedema:** when a lymphatic vessel is compressed, blocked, or its valves are damaged, lymph drainage slows/stops in the affected area. Fluid continues leaving blood capillaries there, but the lymphatic system can no longer remove it — interstitial fluid volume/pressure gradually increase, causing tissue swelling (lymphedema).

Larger Lymphatic Vessels

- Lymph flows from lymphatic capillaries into larger lymphatic vessels leading toward the trunk. Differ histologically from veins: (1) thinner walls; (2) wider lumens; (3) no clear boundaries between the three vessel wall layers.
- Small/medium lymphatic vessels have internal **valves** (like veins) that prevent lymph backflow, especially in the limbs. Pressure within the lymphatic system is very low, so valves are essential to maintain normal lymph flow toward the thoracic cavity. Skeletal muscle contractions help move lymph through the vessels.
- Each lymphatic valve consists of a pair of flaps permitting fluid movement in only one direction (resembles venous valves).

23.3 | Major Lymph-Collecting Vessels

- **Superficial lymphatics** and **deep lymphatics** collect lymph from lymphatic capillaries.
- **Superficial lymphatics:** anastomose freely, more numerous than superficial veins; travel adjacent to superficial veins; found in: (1) the subcutaneous layer deep to the skin, (2) loose connective tissues of mucous membranes lining the digestive, respiratory, urinary, reproductive tracts, (3) loose connective tissues of serous membranes lining the pleural, pericardial, and peritoneal cavities.
- **Deep lymphatics:** large lymphatic vessels accompanying deep arteries and veins; collect lymph from skeletal muscles and other organs of the neck, limbs, trunk, and visceral organs in the thoracic and abdominopelvic cavities.
- Superficial and deep lymphatics converge to form larger vessels called **lymphatic trunks**. The five lymphatic trunks: (1) **lumbar trunks**, (2) **intestinal trunks**, (3) **bronchomediastinal trunks**, (4) **subclavian trunks**, (5) **jugular trunks**.

The Thoracic Duct

- The thoracic duct collects lymph from BOTH sides of the body inferior to the diaphragm AND from the LEFT side of the body superior to the diaphragm.
- Begins as the **cisterna chyli** — an expanded, saclike chamber.
- **Cisterna chyli:** begins inferior to the diaphragm at the level of vertebra **L2**, anterior to the vertebral column, slightly right of midline. Here the deep lumbar trunks and intestinal trunk come together to form a single lymphatic vessel. Receives lymph from the abdomen, pelvis, and lower limbs via the right and left **lumbar trunks** and the **intestinal trunk**.
- As the cisterna chyli passes through the diaphragm with the aorta (at the **aortic hiatus**), it becomes the **thoracic duct**.
- At the level of vertebra **T5**, the thoracic duct shifts left, coming to lie along the left side of the esophagus.
- The thoracic duct ascends to the level of the left clavicle. It collects lymph from the **left bronchomediastinal trunk**, the **left subclavian trunk**, and the **left jugular trunk**.
- The thoracic duct empties into the **left subclavian vein**, near the left internal jugular vein.
- Lymph collected from the left side of the head, neck, and thorax, as well as lymph from the entire body inferior to the diaphragm, re-enters the venous circulation via the thoracic duct.
- Anatomical relations (from the dissection figure): the thoracic duct lies in the posterior thorax alongside the **thoracic aorta**, **azygos vein**, esophagus, and pleura; the cisterna chyli lies near the **abdominal aorta**, inferior to the diaphragm.

The Right Lymphatic Duct (for comparison/contrast)

- Drains the right half of the body superior to the diaphragm (region of the right clavicle — right side of head, neck, thorax, and right upper limb).

- Empties into the venous system at or near the junction of the **right internal jugular vein** and **right subclavian vein**.
- Contrast with thoracic duct: thoracic duct drains both sides of the body inferior to the diaphragm plus the left side superior to the diaphragm (i.e., roughly 3/4 of the body); right lymphatic duct drains only the right side superior to the diaphragm (roughly 1/4 of the body).

23.4 | Lymphocytes (intro, T Cells, B Cells)

- Lymphocytes are responsible for specific immunity. They respond with a combination of physical and chemical attack to: (1) invading organisms (bacteria, viruses), (2) abnormal body cells (virus-infected or cancer cells), (3) foreign proteins (e.g., bacterial toxins).
- Lymphocytes circulate throughout the body in the bloodstream and lymph; exit the bloodstream, move through interstitial space/peripheral tissues, and eventually return to the bloodstream via the lymphatic system. Time spent within the lymphatic system varies — a lymphocyte may remain within a lymphatic organ for hours, days, or even years. When a lymphocyte encounters an invading pathogen or foreign protein, it initiates an immune response.
- Three classes of lymphocytes in the blood (few/no structural differences; classified by biochemical/functional characteristics):
 - **T cells** (thymus-dependent).
 - **B cells** (bone marrow-derived).
 - **NK cells** (natural killer).
- Adaptive (specific) defenses result from the coordinated activities of T cells and B cells (T lymphocytes and B lymphocytes).

T Cells

- Approximately **80%** of circulating lymphocytes are T cells.
- Originate within the red bone marrow but must migrate to the **thymus** to become activated (immunocompetent).
- Under proper stimulation, T cells differentiate into several types:
 - **Cytotoxic T cells:** direct cellular attack, provide cell-mediated immunity; enter peripheral tissues, attack antigens physically and chemically.
 - **Helper T cells:** stimulate responses of both T cells and B cells; must activate B cells before B cells can produce antibodies.
 - **Regulatory T cells:** subset of T cells that moderate the immune response; interplay between helper T cells and regulatory T cells helps establish/control immune response sensitivity. Other regulatory T cell types (e.g., suppressor/inducer T cells) suppress B cell activity but stimulate other T cells.
 - **Memory T cells:** respond to antigens already encountered by cloning (producing identical cellular copies) to ward off the invader; if antigens eliminated in a primary response reappear, a rapid secondary response is initiated by memory cells.

B Cells

- Originate and become immunocompetent within the red bone marrow. Account for **10–15%** of circulating lymphocytes.
- Activated B cells divide, producing daughter cells that differentiate into:
 - **Plasma cells (plasmocytes):** responsible for producing/secreting antibodies.
 - **Memory B cells.**
- **Antibodies** (immunoglobulins): when bound to an antigen, trigger a chain of events leading to destruction, neutralization, or elimination of the antigen.
- B cells provide **antibody-mediated immunity** — a defense within body fluids.

Lymphocytopoiesis (lymphocyte formation) — overview

- Involves the red bone marrow, thymus, and peripheral lymphatic tissues.
- Hematopoietic stem cells in the red bone marrow produce lymphatic stem cells with distinct fates:

- One group remains in the bone marrow, producing daughter cells that mature into **NK cells** and **B cells** — as they mature, B cells and NK cells enter the bloodstream and migrate to peripheral tissues.
- Another group migrates to the **thymus**, producing mature **T cells** — mature T cells enter the bloodstream and migrate to the red bone marrow, spleen, and other lymphatic tissues.
- Functional roles of mature lymphocytes:
 - **Immune surveillance:** NK cells circulate throughout the body, detecting/responding to toxins and pathogens threatening homeostasis; NK cells attack foreign cells, body cells infected by viruses, and cancer cells.
 - **Antibody-mediated immunity:** when stimulated, B cells differentiate into plasma cells, which produce/secrete antibodies; antibodies attach to pathogens, chemically lysing the plasma membrane of abnormal cells → cell destroyed.
 - **Cell-mediated immunity:** cytotoxic T cells (a mature T cell type) attach to abnormal cells or other specific targets; this attachment starts a chain reaction leading to destruction of the target → cell destroyed.

Ch 19 — The Thymus (p. 514–518, within the Endocrine System chapter)

19.5 | The Thymus

- **Location:** the thymus lies just posterior to the sternum, within the **mediastinum** of the thoracic cavity.
- **Size changes with age:**
 - In newborn infants and young children, the thymus is relatively large, extending from the base of the neck to the superior border of the heart.
 - Although its **relative** size (relative to overall body size) decreases as a child grows, the thymus continues to slowly enlarge in absolute terms.
 - The thymus reaches its **maximum size just before puberty**, weighing approximately **40 g**.
 - After puberty, it gradually diminishes in size (involution); by **age 50** the thymus weighs **less than 12 g**.
- **Function:** the thymus produces several hormones important in developing and maintaining immune defenses.
 - **Thymosin:** name originally given to a thymic extract that promoted the development and maturation of lymphocytes (the white blood cells responsible for immunity).
 - Researchers have determined "thymosin" is actually a blend of several different hormones: **thymosin-1, thymopoietin, thymopentin, thymulin, thymic humoral factor, and IGF-1**.
 - It has been suggested (though researchers do not totally agree) that the gradual decrease in size and secretory ability of the thymus with age may make the elderly more susceptible to disease.
 - (The histological organization of the thymus and the functions of the various "thymosins" are discussed in detail in Chapter 23 — the Lymphatic System chapter.)
- **Endocrine summary table entry (Thymus):** Hormone-producing cell type = **epithelial reticular cells**; hormones = **thymosins**; target = **lymphocytes**; effect = **maturation and functional competence of the immune system**.
- Concept check emphasis: be able to describe the anatomical location of the thymus (posterior to the sternum, within the mediastinum).

Ch 17 — Cardiac, Pulmonary, Esophageal Autonomic Plexuses (p. 460–465)

General Functions of the Parasympathetic Division (context)

- Partial listing of major effects produced by the parasympathetic division:
 - Constriction of the pupils (restricts light entering the eyes, aids focusing on nearby objects).
 - Secretion by digestive glands (salivary glands, gastric glands, duodenal/other intestinal glands, pancreas, liver).
 - Secretion of hormones promoting nutrient absorption by peripheral cells.
 - Increased smooth muscle activity along the digestive tract.
 - Stimulation and coordination of defecation.
 - Contraction of the urinary bladder during urination.
 - **Constriction of the respiratory passageways.**
 - **Reduction in heart rate and force of contraction.**
 - Sexual arousal and stimulation of sexual glands in both sexes.
- These functions center on relaxation, food processing, and energy absorption. Parasympathetic stimulation increases nutrient content in the blood; cells throughout the body absorb nutrients to support growth and anabolic activities.
- Preganglionic parasympathetic fibers synapse in the **ciliary, pterygopalatine, submandibular, and otic ganglia** (head structures); short postganglionic fibers continue to peripheral targets.
- The **vagus nerve (X)** provides preganglionic parasympathetic innervation to intramural ganglia within viscera in the thoracic and abdominopelvic cavities, traveling as far as the last segments of the large intestine. **The vagus nerve alone provides roughly 75% of all parasympathetic outflow.**
- Sacral parasympathetic outflow does not join the ventral rami of spinal nerves; instead, preganglionic fibers form distinct **pelvic nerves** that innervate intramural ganglia in the kidney/urinary bladder, terminal large intestine, and sex organs.
- Parasympathetic neuroeffector junctions are small, with narrow synaptic clefts; effects of stimulation are short-lived because most released ACh is inactivated by **acetylcholinesterase (AChE)** within the synapse — so parasympathetic stimulation effects are localized and brief.

Cholinergic Receptor Types (context for understanding plexus signaling)

1. **Nicotinic receptors:** on surfaces of all ganglionic neurons (both parasympathetic and sympathetic divisions) and at neuromuscular synapses of the somatic nervous system. ACh exposure always causes excitation via opening of chemically gated Na⁺ channels on the postsynaptic membrane.

2. **Muscarinic receptors:** found at all cholinergic neuromuscular/neuroglandular junctions in the parasympathetic division, and at the few cholinergic neuroeffector junctions in the sympathetic division. Stimulation produces longer-lasting effects than nicotinic receptor stimulation; response reflects activation/inactivation of specific enzymes and may be excitatory or inhibitory.

- Names derive from the chemicals that stimulate the receptor sites: nicotinic receptors bind nicotine; muscarinic receptors bind muscarine.

Summary of the Parasympathetic Division

- Includes visceral motor nuclei in the brainstem associated with four cranial nerves: **III, VII, IX, and X**. Autonomic nuclei also lie in the lateral portions of the anterior horns in sacral segments **S2–S4**.
- Ganglionic neurons located in **terminal** or **intramural ganglia**.
- Innervates structures in the head and organs in the thoracic and abdominopelvic cavities.

- All parasympathetic neurons are **cholinergic**. ACh release by preganglionic neurons stimulates nicotinic receptors on ganglionic neurons — always excitatory. ACh release at neuroeffector junctions stimulates muscarinic receptors — effects may be excitatory or inhibitory depending on the enzymes activated.
- Effects of parasympathetic stimulation are brief and restricted to specific organs/sites.

17.4 | Relationship between the Sympathetic and Parasympathetic Divisions

- The sympathetic division has widespread impact, reaching visceral organs/tissues throughout the body. The parasympathetic division modifies activity of structures innervated by specific cranial nerves and pelvic nerves — including visceral organs within the thoracic and abdominopelvic cavities.
- Some organs are innervated by only one autonomic division; **most vital organs receive dual innervation** (both sympathetic and parasympathetic).
- Where dual innervation exists, the two divisions often have **opposite (antagonistic) effects**. Dual innervation is most common in the digestive tract, the heart, and the lungs. Example: sympathetic stimulation decreases digestive tract motility, while parasympathetic stimulation increases it.

Anatomy of Dual Innervation

- In the head: parasympathetic postganglionic fibers from the ciliary, pterygopalatine, submandibular, and otic ganglia accompany cranial nerves to peripheral destinations. Sympathetic innervation reaches the same structures by traveling directly from the **superior cervical ganglia** of the sympathetic chain.
- **In the thoracic and abdominopelvic cavities, sympathetic postganglionic fibers intermix with parasympathetic preganglionic fibers at a series of plexuses: the cardiac plexus, pulmonary plexus, esophageal plexus, celiac plexus, inferior mesenteric plexus, and hypogastric plexus.** Nerves leaving these plexuses travel with the blood vessels and lymphatics supplying visceral organs.

The Thoracic Autonomic Plexuses (core content for this unit)

- Autonomic fibers entering the thoracic cavity intersect at the **cardiac plexus** and the **pulmonary plexus**.
 - These plexuses contain BOTH sympathetic fibers innervating the heart AND parasympathetic fibers innervating the heart and lungs.
- The **esophageal plexus** contains descending branches of the **vagus nerve** and **splanchnic nerves** leaving the sympathetic chain ganglia on each side.
- Parasympathetic preganglionic fibers of the vagus nerve follow the esophagus as it enters the abdominopelvic cavity; there they join the network of the **celiac plexus (solar plexus)**.
- The celiac plexus and an associated smaller plexus, the **inferior mesenteric plexus**, innervate viscera within the abdominal cavity.
- The **hypogastric plexus** contains the parasympathetic outflow of the pelvic nerves, sympathetic postganglionic fibers from the inferior mesenteric ganglion, and sacral splanchnic nerves from the sympathetic chain; it innervates the digestive, urinary, and reproductive organs of the pelvic cavity.
- (Diagrammatic summary: ANS plexuses in the thoracic cavity = cardiac, esophageal, pulmonary plexuses; in the abdominopelvic cavity = celiac, inferior mesenteric, hypogastric plexuses.)

Comparison Table: Sympathetic vs. Parasympathetic Divisions

Characteristic	Sympathetic Division	Parasympathetic Division
Location of CNS visceral motor neurons	Lateral horns of spinal segments T1–L2	Brainstem and spinal segments S2–S4
Location of PNS ganglia	Sympathetic chain ganglia; collateral ganglia (celiac, superior mesenteric, inferior mesenteric) anterior/lateral to the descending aorta	Within the tissues of the target organs (intramural ganglion) or close to the target organ (terminal ganglion)
Preganglionic fiber length	Relatively short, myelinated	Relatively long, myelinated
Postganglionic fiber length	Relatively long, unmyelinated	Relatively short, unmyelinated

| Neurotransmitter released | Usually norepinephrine | Always acetylcholine |

| Neuroeffector junction | Varicosities and enlarged terminal knobs releasing transmitter near target cells | Neuroeffector junctions releasing transmitter to a special receptor surface |

| Degree of divergence (CNS to ganglion cells) | Approximately 1:32 | Approximately 1:6 |

| General functions | Stimulate metabolism, increase alertness, prepare for emergency ("fight or flight") | Promote relaxation, nutrient uptake, energy storage ("rest and digest") |

Visceral Reflexes

- **Visceral reflexes:** autonomic reflexes initiated in the viscera; provide automatic motor responses that can be modified, facilitated, or inhibited by higher centers, especially the hypothalamus.
- Example: a light shone in one eye triggers a visceral reflex (pupillary reflex); nuclei directing pupillary constriction/dilation are also controlled by hypothalamic centers concerned with emotional states — e.g., pupils constrict when queasy/nauseated, dilate when sexually aroused.
- Visceral reflex arcs share the same basic components as somatic reflexes (receptor, sensory nerve, etc.), but **all visceral reflexes are polysynaptic**.
- Visceral sensory neurons deliver information to the CNS via the dorsal roots of spinal nerves, sensory branches of cranial nerves, and autonomic nerves innervating visceral effectors.
- Visceral reflexes are either **long reflexes** or **short reflexes**:
 - **Long reflexes:** resemble polysynaptic somatic reflexes; processing involves interneurons within the CNS; motor neurons of these reflexes are located in the brainstem or spinal cord; the ANS carries motor commands to visceral effectors after a synapse within a peripheral autonomic ganglion.
 - **Short reflexes:** bypass the CNS entirely; involve sensory neurons and interneurons with neuronal somas located within autonomic ganglia; interneurons synapse on ganglionic neurons; motor commands distributed by postganglionic fibers; control very simple motor responses with localized effects.
- In general: short reflexes control activity patterns in one part of a target organ; long reflexes coordinate activities of the entire organ.

Table of Important Visceral Reflexes (selected — those most relevant to thorax structures)

- **Coughing reflex** (Ch 24): stimulus = irritation of respiratory tract lining → response = sudden explosive ejection of air → coordinated by the coughing center in the medulla oblongata.
- **Baroreceptor reflex** (Ch 21): stimulus = sudden rise in blood pressure in the carotid artery → response = reduction in heart rate and force of contraction → coordinated in the cardiac center in the medulla oblongata.
- **Cardioacceleratory reflex** (sympathetic) (Ch 21): stimulus = sudden decline in blood pressure in the carotid artery → response = increase in heart rate and force of contraction → coordinated in the cardiac center in the medulla oblongata.
- **Vasomotor reflexes** (sympathetic) (Ch 22): stimulus = changes in blood pressure in major arteries → response = changes in diameter of peripheral blood vessels → coordinated in the vasomotor center in the medulla oblongata.
- **Swallowing reflex** (Ch 25): stimulus = movement of food/drink into the superior pharynx → response = smooth and skeletal muscle contractions → coordinated by the swallowing center in the medulla oblongata.
- **Vomiting reflex** (Ch 25): stimulus = irritation of digestive tract lining → response = reversal of normal smooth muscle action to eject contents → coordinated by the vomiting center in the medulla oblongata.
- **Direct light and consensual light reflexes** (Ch 18): stimulus = bright light shining in eye(s) → response = constriction of pupils of both eyes → mediated by the oculomotor nerve (III).
- **Pupillary reflex** (sympathetic) (Ch 18): stimulus = low light level reaching visual receptors → response = dilation of pupil → mediated by the oculomotor nerve (III).
- (Also listed in the full table but pertaining to other regions: gastric/intestinal reflexes mediated by the vagus nerve; defecation and urination reflexes requiring voluntary sphincter relaxation; sexual arousal/emission/ejaculation reflexes.)

Clinical Note — Urinary Bladder Dysfunction Following Spinal Cord Injury (SCI)

(included as it appears within this page range as a worked clinical example of visceral reflex disruption, though the organ itself is pelvic rather than thoracic)

- Normally, when the bladder is full, coordinated visceral (autonomic) reflexive contraction of the detrusor muscle and relaxation of the involuntary internal urethral sphincter are accompanied by voluntary (somatic) relaxation of the external urethral sphincter.
- Afferent sensory nerves from bladder wall stretch receptors enter the spinal cord between **L1 and S4**. Efferent parasympathetic nerves **S2–S4** innervate the detrusor and internal involuntary sphincter — increased parasympathetic activity causes bladder contraction and urethral sphincter relaxation (urge to urinate).
- Efferent sympathetic nerves **L1 and L2** inhibit bladder contraction and cause the involuntary urethral sphincter to contract, preventing urine passage.
- **Pudendal nerves (S2–S4)** of the somatic nervous system regulate relaxation of the voluntary skeletal muscle of the external urethral sphincter, allowing urination.
- After an SCI, the kidneys are unaffected, but messages can no longer move between the bladder and brain. During spinal shock (days to weeks), there is no somatic or visceral bladder activity — the bladder is completely flaccid, requiring continuous catheter drainage.
- After spinal shock subsides, bladder/sphincter function depends on injury level: injury at **T12 or above** → reflexive, spastic bladder that empties without control. Disruption of the reflex arc in spinal nerves **S2–S4** → flaccid bladder that cannot contract and therefore cannot empty.
- Involuntary urination = **incontinence**. Maintaining urinary tract health post-SCI requires an indwelling urinary catheter or intermittent catheterization.

Ch 6.8 — The Thoracic Cage (p. 167–171)

6.8 | The Thoracic Cage — Overview

- The thoracic cage serves two functions:
 1. **Protects** the heart, lungs, thymus, and other structures in the thoracic cavity.
 2. **Serves as an attachment point** for muscles involved with: (1) breathing, (2) maintaining the position of the vertebral column, and (3) movements of the pectoral girdle (clavicles and scapulae) and upper limbs.
- The skeleton of the thoracic cage consists of the **thoracic vertebrae, ribs, costal cartilages, and sternum**. The ribs and sternum together form the **rib cage**.

The Ribs

- There are **12 pairs of ribs**, classified into three categories:
 - **True ribs (vertebrosternal ribs)**: the upper **7 pairs** (ribs 1–7). Anteriorly connected to the sternum by separate cartilages, the **costal cartilages**. Beginning with rib 1, the vertebrosteral ribs gradually increase in length and radius of curvature.
 - **False ribs (vertebrochondral ribs)**: **ribs 8–10**. Do not attach directly to the sternum — their costal cartilages fuse together before reaching the sternum (rib pairs 8–10 attach to the sternal body via a single shared pair of cartilages with rib pair 7).
 - **Floating ribs**: **ribs 11–12** (the last two pairs). Have no connection with the sternum at all.
- **Rib articulation with the vertebral column**:
 - The **head** of each rib articulates with the body of a thoracic vertebra or between adjacent vertebral bodies.
 - After a short **neck**, the **tubercle** projects posteriorly. The inferior portion of the tubercle contains an articular facet that contacts the transverse process of thoracic vertebrae T1–T11.
 - When a rib articulates between adjacent vertebrae, the **interarticular crest** divides the articular surface into superior and inferior articular facets.

- **Ribs 1–10** originate at costal facets on the bodies of vertebrae T1–T10 (per Concept Check reference, articular facets articulate with the transverse costal facets of their respective vertebrae).
- **Ribs 11 and 12** originate at costal facets on **T11 and T12**. These ribs do NOT have tubercular facets and do NOT articulate with transverse processes (a key structural difference from ribs 1–10).
- The **bend/angle** of the rib marks where the tubular **body (shaft)** begins curving toward the sternum.
- The internal rib surface is **concave**; the **costal groove** on the inferior border marks the path of the **intercostal nerves and blood vessels**.
- The superficial rib surface is **convex** and serves as the attachment site for muscles of the pectoral girdle and trunk. The **intercostal muscles** that move the ribs attach to the superior and inferior rib surfaces.
- **Rib mobility**: with complex musculature, dual articulations at the vertebrae, and a flexible connection to the sternum, the ribs are quite mobile.
 - Ribs curve away from the vertebral column, angling downward.
 - A typical rib functions like the **handle on a bucket**, lying just below the horizontal plane: pushing it down forces it inward; pulling it up swings it outward.
 - Because of rib curvature, these same movements also change the sternum's position: **depressing** the ribs moves the sternum **posteriorly (inward)**; **elevating** the ribs moves it **anteriorly (outward)**.
 - As a result, rib movements affect both the **width** and **depth** of the thoracic cage, increasing or decreasing its volume accordingly (this is the anatomical basis for the "bucket-handle" mechanism of breathing).

The Sternum

- The adult **sternum** is a flat bone in the anterior midline of the thoracic wall, consisting of three parts: **manubrium, body, xiphoid process**.
- **Manubrium**:
 - Articulates with the **clavicles** (collarbones) of the appendicular skeleton and the costal cartilages of the **first pair of ribs**.
 - The widest and most superior portion of the sternum.
 - The **jugular notch**: shallow indentation on the superior surface of the manubrium, located between the clavicular articulations.
- **Body**:
 - Attaches to the inferior surface of the manubrium, extends caudally along the midline.
 - Individual costal cartilages from rib pairs **2–7** attach to this portion.
 - Rib pairs **8–10** also attach to the body, but via a single pair of cartilages shared with rib pair 7.
- **Xiphoid process**:
 - The smallest part of the sternum; attached to the inferior surface of the body.
 - The muscular **diaphragm** (breathing muscle) and the **rectus abdominis** muscle attach to the xiphoid process.
- **Ossification**: begins in **6 to 10 different ossification centers**; fusion is completed by **age 25 or later**. Before age 25, the sternal body consists of **four separate bones**, whose boundaries are visible as transverse lines crossing the adult sternum. The xiphoid process is the last component of the sternum to undergo ossification and fusion.
- **Clinical correlation**: the xiphoid process's connection to the sternal body can be broken by impact or strong pressure, creating a bone "spear" that can severely damage the liver. To reduce this risk, strong emphasis is placed on proper hand positioning during **CPR (cardiopulmonary resuscitation)** training.
- Concept-check emphasis: improper CPR administration could fracture the **xiphoid process**; the costal groove houses the **intercostal nerves and blood vessels**.

Ch 10 — Thoracic Wall / Respiratory Muscles (p. 259–275)

10.1 | The Four Groups (Axial Musculature Overview)

- The muscular system, like the skeletal system, is divided into **axial** and **appendicular** divisions. The axial musculature originates on the axial skeleton; positions the head and vertebral column; helps breathing by moving the rib cage. The appendicular musculature inserts onto and stabilizes/moves the appendicular skeleton.
- Axial muscles are organized into **four groups**:

1. **Muscles of the head and neck**: responsible for verbal/nonverbal communication (laughing, talking, frowning, smiling, whistling); also involved in chewing, swallowing, moving the eyes. *(Detailed coverage of this group — facial expression, mastication, tongue, pharynx, hyoid muscles — belongs to the Head/Neck unit and is not detailed here.)*

2. **Muscles of the vertebral column**: includes flexors and extensors of the axial skeleton.

3. **Oblique and rectus muscles**: form the walls of the thoracic and abdominal cavities and (in the neck) also include oblique/rectus muscles, even though these don't form a muscular wall there — included in this group because of shared embryological origin with the other chest wall muscles. In the thoracic area, the ribs separate these muscles; over the abdominal surface, they form broad muscular sheets. *(This is the group containing the intercostals, transversus thoracis, abdominal wall muscles, and — grouped here due to shared embryological origin — the diaphragm. This is the core "thoracic wall / respiratory muscle" content.)*

4. **Muscles of the perineal region and pelvic diaphragm**: extend between the sacrum and pelvic girdle; support pelvic cavity organs, flex joints of the sacrum/coccyx, control the urethra and anus. *(Pelvic unit content, not detailed here.)*

10.3 | Muscles of the Vertebral Column (brief — back musculature context)

- Muscles of the vertebral column are covered by more superficial back muscles; include many posterior extensors but few anterior flexors.
- Divided into **extrinsic back muscles** (associated with upper extremity/shoulder movement) and **intrinsic back muscles** (fully contained — origin, belly, insertion — within, and acting upon, the back).
- Intrinsic (true) back muscles arranged into **superficial, intermediate, and deep layers**, found lateral to the vertebral column, between the spinous and transverse processes.

Superficial layer — the Erector Spinae group

- **Splenius muscles**: most superficial intrinsic back muscles. Splenius capitis originates on the ligamentum nuchae and spines of C7–T4, inserts on the skull. Splenius cervicis originates on the ligamentum nuchae and spines of T3–T6, inserts onto C1–C3. Both extend and laterally flex the neck.
- **Erector spinae** subdivided into three groups (medial to lateral): **spinalis, longissimus, iliocostalis**.
 - Spinalis = most medial group; iliocostalis = most lateral group.
 - Bilateral contraction → extends the vertebral column; unilateral contraction → laterally flexes and rotates the vertebral column to the ipsilateral side.
 - Naming convention: "capitis" = inserts on the skull; "cervicis" = inserts on upper cervical vertebrae; "thoracis"/"lumborum" = thoracic/lumbar regions.
 - **Iliocostalis group** (most lateral; rib-related actions relevant to thorax):
 - Iliocostalis cervicis: superior borders of vertebrosteral ribs near the angles → transverse processes of C4–C6; extends/laterally flexes neck, **elevates ribs**.
 - Iliocostalis thoracis: superior borders of ribs 6–12 medial to the angles → superior ribs and transverse process of C7; stabilizes thoracic vertebrae in extension.
 - Iliocostalis lumborum: iliac crest, sacral crests, lumbar spinous processes → inferior surfaces of ribs 6–12 near their angles; extends vertebral column, **depresses ribs**.

Deep layer — Transversospinales muscles

- Deepest layer of true back muscles; interconnects and stabilizes the vertebrae. Consists of: **semispinalis, multifidus, rotatores, interspinales, intertransversarii**.
- Short muscles working in combination to produce limited extension and lateral flexion of the vertebral column; also rotate the vertebral column contralaterally; important in fine adjustments of individual vertebral positions.
- If injured, these muscles can start a pain → muscle stimulation → contraction → pain cycle, leading to pressure on adjacent spinal nerves, sensory losses, limited mobility (rationale for warm-up/stretching before exercise).

Spinal Flexors

- Few spinal flexors exist (compared to numerous extensors) partly because many large trunk muscles flex the vertebral column when they contract, and because most body weight lies anterior to the vertebral column, so gravity tends to flex the spine.
- In the neck: **longus capitis** and **longus colli** rotate and flex the neck (depending on unilateral/bilateral contraction).
- In the lumbar region: the large **quadratus lumborum** laterally flexes the vertebral column and **depresses the ribs**.

10.4 | Muscles of the Oblique and Rectus Groups (Thoracic Wall & Respiratory Muscles — core content)

- Key Point: the oblique muscles compress underlying structures and rotate the vertebral column. The rectus muscles are important flexors of the vertebral column and are antagonists to the erector spinae muscles.
- The oblique and rectus muscles lie between the vertebral column and the anterior midline; divided into three regional groups: **cervical, thoracic, and abdominal**. (Grouped together with the diaphragm due to shared embryological origin.)

Oblique muscles of the neck — Scalenes

- The oblique muscles include the **scalenes** of the neck and the **intercostal** and **transversus** muscles of the thorax.
- **Anterior, middle, and posterior scalenes:** elevate the first two ribs and flex the neck.
 - Scalenus anterior: transverse and costal processes of C3–C6 → superior surface of first rib; elevates ribs and/or flexes neck; unilateral action bends neck and rotates.
 - Scalenus middle: transverse and costal processes of the atlas (C1) and C3–C7 → superior surface of first rib; same actions.
 - Scalenus posterior: transverse and costal processes of C4–C6 → superior surface of second rib; same actions.
 - Innervation: cervical spinal nerves.

Oblique muscles of the thorax — Intercostals and Transversus Thoracis

- In the thorax, the oblique muscles lie between the ribs and are called **intercostal muscles**. The **external intercostals** cover the **internal intercostals**.
- Both sets of intercostal muscles aid in breathing movements of the ribs.
- **External intercostals:** origin = inferior border of each rib; insertion = superior border of the more inferior rib; action = **elevate ribs**; innervation = intercostal nerves (branches of thoracic spinal nerves).
- **Internal intercostals:** origin = superior border of each rib; insertion = inferior border of the more superior rib; action = **depress ribs**; innervation = intercostal nerves.
- **Transversus thoracis:** a small muscle crossing the inner surface of the rib cage, separated from the pleural cavity by the parietal pleura (a serous membrane). Origin = posterior surface of sternum; insertion = cartilages of ribs; action = **depress ribs**; innervation = intercostal nerves.
- (Also in this thoracic region: **Serratus posterior superior** — origin = spinous processes of C7–T3; insertion = superior borders of ribs 2–5; action = elevate ribs, enlarge thoracic cavity; innervation = thoracic nerves. **Serratus posterior inferior** — origin = aponeurosis from spinous processes of T9–L2 or L3; insertion =

inferior borders of ribs 9–12; action = pull ribs inferiorly and outward, opposing the diaphragm; innervation = thoracic nerves.)

Oblique muscles of the abdomen

- The abdominal oblique muscles show the same three-layer muscular pattern as the thorax, each layer arranged similarly to the thoracic cavity layers: **external oblique**, **internal oblique**, and **transversus abdominis**. This three-layer arrangement strengthens the abdominal wall.
 - **External oblique:** origin = external/inferior borders of ribs 5–12; insertion = aponeuroses extending to the linea alba and iliac crest; action = compresses abdomen, depresses ribs, flexes/laterally flexes/rotates the vertebral column to the opposite side; innervation = intercostal nerves 5–12, iliohypogastric, ilioinguinal nerves.
 - **Internal oblique:** origin = thoracolumbar fascia, inguinal ligament, iliac crest; insertion = inferior surfaces of ribs 9–12, costal cartilages 8–10, linea alba, pubis; action = same as external oblique but rotates the vertebral column to the SAME side; innervation = intercostal nerves 5–12, iliohypogastric, ilioinguinal nerves.
 - **Transversus abdominis:** origin = cartilages of ribs 7–12, iliac crest; insertion = linea alba and pubis; action = compresses abdomen; innervation = intercostal nerves 5–12, iliohypogastric, ilioinguinal nerves.

Rectus muscle of the abdomen

- **Rectus abdominis:** originates at the xiphoid process (and cartilages of ribs 5–7), inserts onto the pubic bone (superior surface of pubis around the symphysis). Action = depress ribs, flex vertebral column, and compress abdomen. Innervation = intercostal nerves (T7–T12).
- The **linea alba** ("white line") — a band of fibrous connective tissue — divides the rectus abdominis longitudinally.
- **Transverse tendinous inscriptions:** bands of fibrous connective tissue that divide the rectus abdominis into four repeated segments (creating the "six-pack" appearance).

The Diaphragm

- Key Point: the diaphragm is a **dome-shaped sheet of skeletal muscle** separating the thoracic and abdominal cavities.
- **Contracting** the diaphragm **flattens** the muscle, **increasing the volume of the thoracic cavity** and causing **inhalation**.
- **Relaxing** the diaphragm **decreases** the volume of the thoracic cavity, aiding **exhalation**.
- Muscle table data: origin = xiphoid process, ribs 7–12 and associated costal cartilages; insertion = central tendon sheet; action = contraction expands the thoracic cavity and compresses the abdominopelvic cavity; innervation = **phrenic nerves (C3–C5)**.
- The diaphragm is grouped with the oblique/rectus muscles in this textbook's organizational scheme because of their shared embryological origin.
- (Text notes that the muscles of breathing are examined in further detail in Chapter 24 — the Respiratory System chapter.)
- Concept-check emphasis: damage to the external intercostal muscles interferes with normal breathing/inspiration (rib elevation).

Ch 12 — Cross-Sectional Anatomy: T2, T8, T10 (p. 335–337)

Cross Section at the Level of Vertebra C2 (Figure 12.9) — context

- Level: head, at the level of the axis (C2).
- **Anterior structures:** orbicularis oris, maxilla, median lingual raphe, buccinator, masseter, body of C2 (axis), ramus of mandible, pterygoid muscles, longus capitis.
- **Vessels/nerves:** internal carotid artery, internal jugular vein, vertebral artery, spinal cord.

- **Muscles:** sternocleidomastoid, obliquus capitis inferior, longissimus capitis, splenius, rectus capitis posterior major, semispinalis capitis (lateral part and medial part), trapezius.
- Orientation: anterior structures listed first, posterior structures (deep back/neck muscles) listed toward the posterior aspect of the section.

Cross Section at the Level of Vertebra T2 (Figure 12.10)

- **Anterior structures:** trachea, esophagus, sternocleidomastoid (sternal head), sternothyroid, clavicle, common carotid artery, subclavius, pectoralis major, pectoralis minor, body of T2, subclavian artery.
- **Shoulder region:** shoulder joint, humerus, subscapularis, scapula, deltoid, infraspinatus.
- **Thoracic viscera:** left lung.
- **Posterior structures:** spinal cord, rhomboid major, splenius cervicis, multifidus, trapezius.
- Clinical/anatomical relevance: this level shows the transition from neck to thorax — trachea and esophagus are anterior to the vertebral body, with the great vessels of the neck (common carotid, subclavian arteries) and the apex of the lung visible, along with the shoulder girdle structures (clavicle, scapula, humerus) at this level.

Cross Section at the Level of Vertebra T8 (Figure 12.11)

- **Anterior structures:** right AV (tricuspid) valve, body of sternum, right atrium, pectoralis major, right lung (middle lobe), rib 4.
- **Heart structures:** right atrium, right ventricle, left ventricle, interventricular septum, right AV (tricuspid) valve — this section passes directly through the heart chambers.
- **Lung structures:** right lung (middle lobe), oblique fissure of right lung, left lung (superior lobe), oblique fissure of left lung, left lung (inferior lobe), right lung (general).
- **Other structures:** esophagus, thoracic aorta, ribs 7 and 8.
- **Posterior structures:** spinal cord, latissimus dorsi, spinous process of T8, trapezius.
- Key teaching point: T8 is approximately the level where a cross section passes through all four heart chambers and both the right and left AV valve region, plus both lungs' oblique fissures — useful landmark for identifying the heart in cross section (interventricular septum divides right ventricle from left ventricle; right atrium and right ventricle are anterior/right, left ventricle is more posterior/left and forms the more muscular chamber).

Cross Section at the Level of Vertebra T10 (Figure 12.12)

- **Anterior structures:** xiphoid process, right lobe of liver, cardiac orifice of the stomach, cardia of the stomach, inferior vena cava, esophagus, azygos vein, thoracic aorta.
- **Vertebral level:** body of T10.
- **Spinal cord:** sacral segments of spinal cord are located at this vertebral level (reflecting the mismatch between spinal cord segment level and vertebral column level, since the cord is shorter than the vertebral column).
- **Abdominal/thoracic viscera at this transitional level:** spleen, diaphragm, left lung (inferior lobe), right lung (inferior lobe) — this section captures the diaphragm and the transition zone between thoracic and abdominal cavities, showing both inferior lung lobes above the diaphragm and upper abdominal organs (liver, stomach cardia, spleen) below/at the diaphragm.
- **Posterior structures:** latissimus dorsi, multifidus, longissimus thoracis, trapezius.
- Key teaching point: T10 is the level of the **esophageal hiatus** (diaphragm opening for the esophagus) in standard anatomical references — this section shows the esophagus, IVC, and thoracic aorta all near the diaphragm, along with the liver, stomach (cardia), and spleen appearing just as abdominal organs come into view.

(For reference — adjacent sections in this range, T12 and L5, are primarily Abdomen/Pelvis unit content)

- **T12** (Figure 12.13): transverse colon, rectus abdominis, transversus abdominis, jejunum, costal cartilage, intercostal muscles of rib 8, ribs 9, ascending colon, right lobe of liver, descending colon, abdominal aorta, renal pelvis of right kidney, renal vein, diaphragm, renal artery, T12–L1 intervertebral disc, left kidney, latissimus dorsi, psoas major, quadratus lumborum, spinal cord, iliocostalis lumborum, spinalis thoracis, longissimus thoracis.
- **L5** (Figure 12.14): ileum, rectus abdominis, descending colon, external oblique, cecum, internal oblique, transversus abdominis, psoas (major), sacrum, iliacus, sacro-iliac joint, ilium, vertebral foramen, ala of sacrum, spinous process of L5, gluteus medius, longissimus thoracis, gluteus maximus.

Clinical Case Wrap-Up — "Breathing Through Your Neck" (Emergency Airway Access)

- Scenario involves an emergency **cricothyroidotomy** (also called cricothyrotomy): an emergency surgical airway created through the **median cricothyroid ligament** (the soft ligamentous space between the thyroid cartilage and cricoid cartilage).
- This space is chosen for emergency "blind" incision because: (1) it is surrounded by a "cartilage cage" (thyroid cartilage above, cricoid cartilage below) that helps guide the incision safely, and (2) there are few blood vessels and nerves crossing this space, and (3) it is safely superior to the thyroid gland.
- Contrast with elective **tracheostomy**: a surgical opening into the trachea itself, made lower and closer to the sternal notch — a more convenient (but less emergency-safe/more vascular) location, typically performed with anesthesia under sterile conditions (not a "blind" emergency procedure).
- Procedure steps described: palpate the anterior neck to find the thyroid cartilage and cricoid cartilage, locate the median cricothyroid ligament between them; make a one-inch horizontal incision directly over the cricothyroid membrane; insert index finger into the wound to feel the "soft spot"; make a horizontal incision into the trachea; insert a tube (in the case description, a straw) about two inches into the airway.
- Concept-check questions posed: (1) In which cervical triangle of the neck is a tracheostomy performed? (2) Where is the best place to practice feeling the median cricothyroid ligament?

Ch 1 — Mediastinum, Pleural & Pericardial Cavities (p. 16–24)

Sectional Anatomy (context — planes of section)

- A **sectional view** is sometimes the only way to illustrate relationships between parts of a three-dimensional object. A slice through a 3D object is described by referencing one of three sectional planes: **frontal, sagittal, or transverse**.
- **Frontal plane** (coronal plane): parallels the longitudinal axis of the body; extends side to side, dividing the body into **anterior and posterior** sections. "Coronal" usually refers to sections passing through the skull.
- **Sagittal plane**: also parallels the longitudinal axis; extends anterior to posterior, dividing the body into **left and right** sections.
 - A section passing along the midline, dividing the body into roughly equal left/right halves = **midsagittal section** (median sagittal section).
 - A section parallel to the midsagittal line but missing the midline = **parasagittal section** (separates right and left portions of unequal size).
- **Transverse plane** (horizontal or cross-sectional plane): lies at right angles to the longitudinal axis of the body part being studied. A division along this plane = **transverse section** (horizontal or cross section).
- **Serial reconstruction**: an important method for studying histological structure and analyzing images from clinical procedures (e.g., following the path of a bent tube like a piece of elbow macaroni — sectional views

change as the plane approaches a curve). Sectional views of internal organs (CT/MRI) can vary widely depending on where the section is taken.

- Convention: when anatomical diagrams/scans present cross-sectional views, they are presented as though the observer is standing at the feet of a person in the supine position, looking toward the head (i.e., viewed from an inferior perspective).

Body Cavities — Overview

- The human body is not a solid object; many organs are suspended in internal chambers called **body cavities** that protect and cushion them.
- The **ventral body cavity** contains organs of the respiratory, cardiovascular, digestive, urinary, and reproductive systems. It is subdivided into the **thoracic cavity** and the **abdominopelvic cavity**; the **diaphragm** (a dome-shaped sheet of skeletal muscle) separates them.
- Internal organs that project into these cavities are called **viscera**. Many organs within these cavities change size/shape as they perform their functions (e.g., the stomach swells at each meal; the heart contracts and expands with each beat). These organs project into moist internal chambers that allow expansion/limited movement while preventing friction.
- Footnote: in the human adult, the thoracic, abdominal, and pelvic cavities share a common embryological origin. The term "dorsal body cavity" (sometimes used for the internal chamber of the skull and the space enclosed by the vertebral arches) refers to chambers defined by bony structures that are anatomically/embryologically distinct from true body cavities — this term is not used in clinical or comparative anatomy.
- Body cavities protect delicate organs and permit changes in the size and shape of visceral organs.
- The diaphragm separates the superior thoracic cavity from the inferior abdominopelvic cavity.
- The thoracic and abdominopelvic cavities contain narrow, fluid-filled spaces lined by a **serous membrane**.

The Thoracic Cavity

- Contains organs of the **respiratory, cardiovascular, and lymphatic systems**, as well as the **thymus** and inferior portions of the **esophagus**.
- Boundaries: the muscles and bones of the chest wall and the diaphragm.
- Subdivided into the **left and right pleural cavities**, separated by the **mediastinum**.

Pleural Cavities and Pleura

- Each pleural cavity contains a lung.
- A shiny, slippery serous membrane called a **pleura** lines each pleural cavity, reducing friction as the lung expands/recoils during breathing.
 - **Visceral pleura**: covers the outer surfaces of each lung.
 - **Parietal pleura**: covers the opposing mediastinal surface and the inner body wall.

The Mediastinum

- The **mediastinum**: connective tissue that surrounds, stabilizes, and supports the **esophagus, trachea, thymus, and major blood vessels** that originate or end at the heart.
- The mediastinum also contains the **pericardial cavity** — a small chamber that surrounds the heart.
- The mediastinum contains: thymus, trachea, esophagus, blood vessels, and the pericardial cavity (which surrounds the heart).

The Pericardium and Pericardial Cavity

- The serous membrane covering the heart is called the **pericardium** (peri-, around + kardia, heart).
- Analogy to visualize the heart/pericardial cavity relationship: a fist pushing into a balloon — the wrist corresponds to the base (attached portion) of the heart, and the balloon corresponds to the pericardium. (The heart projects into the pericardial cavity like a fist pushed into a balloon.)

- The pericardium is composed of two parts:
 - **Parietal layer of the serous pericardium:** an outer sac of tough, fibrous connective tissue.
 - **Visceral layer of the serous pericardium:** inner layer.
- During each heartbeat, the heart changes in size and shape. The **pericardial cavity** permits these changes, and the slippery pericardial lining prevents friction between the heart and adjacent mediastinal structures.
- The membrane lining the pleural cavities is called the **pleura**; the membrane lining the pericardial cavity is called the **serous pericardium**.

The Abdominopelvic Cavity (brief context/contrast)

- Divided into: (1) a superior **abdominal cavity**, (2) an inferior **pelvic cavity**, and (3) an internal chamber called the **peritoneal cavity**.
- The **abdominal cavity** extends from the inferior surface of the diaphragm to an imaginary plane extending from the inferior surface of the lowest (most inferior) spinal vertebra to the anterior and superior margins of the pelvic girdle. Contains the liver, stomach, spleen, kidneys, pancreas, small intestine, and most of the large intestine.
- These organs project partially or completely into the peritoneal cavity, much as the heart and lungs project into the pericardial and pleural cavities, respectively (direct structural parallel to the thoracic serous cavities).
- The **pelvic cavity** (inferior portion of the abdominopelvic cavity) is enclosed by the bones of the pelvis; contains the last segments of the large intestine, the urinary bladder, and reproductive organs (ovaries, uterine tubes, uterus in females; prostate and seminal glands in males). The inferior portion of the peritoneal cavity extends into the pelvic cavity.
- **Peritoneum:** the parietal peritoneum lines the body wall; a narrow, fluid-filled space separates the parietal peritoneum from the visceral peritoneum, which covers the enclosed organs.
- **Mesenteries:** double sheets of peritoneum that suspend organs such as the stomach, small intestine, and portions of the large intestine within the peritoneal cavity — provide blood supply, support, lubrication, and stability while permitting limited movement.

Clinical Note — Pericarditis and Peritonitis

- The suffix **-itis** means "inflammation."
- **Pericarditis:** inflammation of the pericardium; can be caused by any disease-causing agent or trauma; can severely restrict heart function.
- **Peritonitis:** inflammation of the peritoneum (the serous membrane lining the abdomen); may result from bacterial infection, liver failure, kidney failure, or many other causes. Peritonitis affects all organs within the peritoneal cavity.

Clinical Note — Radiological/Imaging Procedures (pp. 20–21)

- **X-rays:** high-energy radiation that penetrates living tissue; a beam travels through the body and strikes a photographic plate; some x-rays are absorbed/deflected passing through the body. **Radiodensity** = resistance to x-ray penetration. Order of increasing radiodensity in the human body: **air, fat, liver, blood, muscle, bone**. Radiodense tissues (e.g., bone) appear white; less dense tissues appear in shades of gray to black. Physicians specializing in performing/analyzing these images are **radiologists**.
- **Barium-contrast x-ray:** barium is very radiodense; used to outline the gastric and intestinal lining against the white of the barium solution (e.g., upper digestive tract imaging).
- **Digital subtraction angiography (DSA):** monitors blood flow through specific organs (brain, heart, lungs, kidneys); x-rays taken before and after radiopaque dye administration; a computer "subtracts" details common to both images, producing a high-contrast image showing dye distribution.
- **CT scan** (computerized tomography; formerly CAT = computerized axial tomography): uses a single x-ray source rotating around the body; the x-ray beam strikes a sensor monitored by a computer; the source completes one revolution every few seconds, then moves and repeats; the computer reconstructs a three-dimensional structure from the rotational data; result is typically a black-and-white sectional view (can be colorized).

- **Spiral CT scan** (helical CT scan): newer 3D imaging technology; the patient advances at a steady pace through the scanner while the imaging source (usually x-rays) rotates continuously around the patient. Because the x-ray detector gathers data quickly and continuously, image quality is higher and patient radiation exposure is lower compared to a standard CT scanner (which collects data more slowly, one slice at a time).
- **MRI (magnetic resonance imaging)**: surrounds part or all of the body with a magnetic field about **3000 times as strong as Earth's**. This field aligns protons within atomic nuclei throughout the body along the magnetic lines of force (like compass needles in Earth's magnetic field). When struck by a radio wave of the proper frequency, a proton absorbs energy; when the pulse ends, that energy is released and detected by the MRI computers.
- **Ultrasound**: a small transmitter contacting the skin broadcasts a brief, narrow burst of high-frequency sound, then picks up the echoes; sound waves reflected by internal structures are used to assemble a picture (**echogram**) from the echo pattern. Images lack the clarity of other procedures, but no adverse effects have been attributed to the sound waves, so fetal development can be monitored without significant risk of birth defects.

Concept Check Emphasis (from this section)

- Know that: an incision just inferior to the diaphragm opens the **abdominopelvic cavity**.
- Know the general function of the mesenteries (blood supply, support, lubrication, stability, permitting limited movement).
- Know what type of section would separate the two eyes (a **sagittal** section, e.g., parasagittal or midsagittal depending on exact placement).